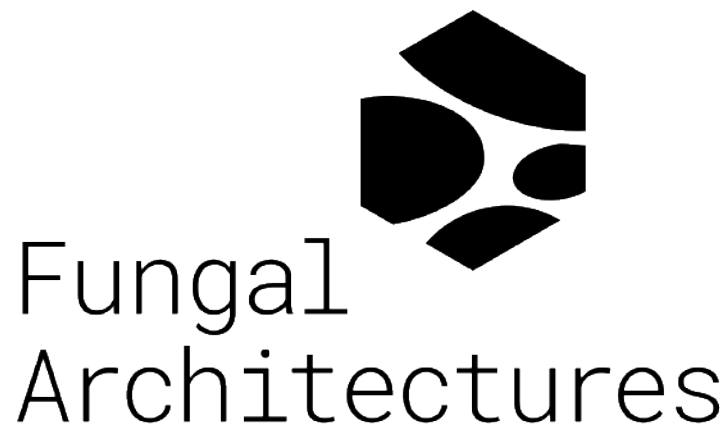


Horizon 2020



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Report on preliminary characterisation and modelling

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Description: In this deliverable, we report on activities and results related to T5.2 *Preliminary Characterisation and Modelling* as defined in the scope of works. This includes preliminary characterisation of mechanical properties of mycelium-based composites (MBC) in compression, tension and bending according to relevant standard tests. Characterisation and modelling of mechanical properties of Kagome weaves is also reported, with particular attention on examining different material compositions, including hybrids of synthetic and organic materials. Thermal characterisation is also conducted on MBC panel materials, and then extended to make preliminary characterisations of a notional cavity wall construction. The mechanical and thermal characterisation work presented here is contextualised against the state-of-the-art, with novel contributions clearly identified. These preliminary characterisation results provide the basis for numerical models which will be integrated in digital design workflows (T5.3 *Design rules for fungal architecture*) to facilitate the specification and prediction of behaviour of novel architectural designs from MBC, and setting the stage for the construction of an ambitious 1:1 demonstrator (T5.4 *Persistent modelling*).

Partners owning: **CITA**Partners contributed: **CITA, Mogu**

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1 INTRODUCTION

Mycelium-Based Composites (MBC) are being investigated in design and materials engineering by leveraging the saprotrophic lifestyle of ligninolytic fungi, taking inspiration in the XIXth to early XXth century method of fungal strain transfer by lignocellulosic solid-state cultivation [34]. Because MBC cultivation protocols can be based on virtually any substrate containing organic polymers such as lignin, hemicellulose, and cellulose, and as they instrumentalise a range of widely available basidiomycota, this class of composite shows potential in obtaining viable products for a variety of uses. Furthermore, MBC conform to circular economy production principles, appear to be biodegradable [118], and are assumed to have a low environmental impact in regards to Life-Cycle Assessment (LCA) – although no peer-reviewed studies have been published to date on this aspect. Lignocellulosic substrates cover a variety of aggregate geometries and chemical profiles, from industrial grade dusts and particles to supplies of irregular shavings, from grain husks to straws; this variety of supplies has led to the emergence of a rich craft in MBC production. However, this poses a challenge in systematically understanding the behaviour of this new class of materials. We argue that rationalising and systematising approaches to analysing their complexity is necessary to actualise their potential and facilitate market readiness.

As composites, MBC are made of a matrix phase that corresponds to a continuous and foam-like or elastomer-like mycelium, and a dispersed phase composed of particles and/or fibres. Additionally, surfacic or volumetric components can be used to compose the substrate of a composite. Fungal decay is functionalised as a binding method between elements of the dispersed phase. From this generic definition we can appreciate the sheer breadth of the design space, and we can identify the main material characteristics for defining the cultivation protocol and specify composite properties:

- Fungal species enzyme array,
- Substrate chemical profile,
- Substrate aggregate characteristics,
- Substrate density and composition,
- Substrate wetting characteristics.

The resulting composite performances depend on the solid-state fermentation (SSF) model that is adopted. While tacit knowledge of MBC craft is sufficient for reaching a satisfactory output, the explicitation and modelling of the driving parameters of an MBC system can foster an engineering practice for it, thus extending the reach and effectiveness of material research and improving market readiness. 3D printing strategies of substrate forming for fungal colonisation are not studied in this report as this technique was not investigated in the context of Fungal Architectures. The inner structure of a typical MBC is presented in Fig.1.

We start by presenting a review of the MBC model from ligninolytic fungi enzymatic activities perspective (section 2.1), and proceed to describe the biochemistry of lignocellulosic substrates while correlating it to fungi enzymatic abilities (section 2.5). Building upon studies of fungal niches in the wild and their parasitic or saprotrophic behaviour in trees, we come to understand the hydrolitic abilities of wood in relation to their chemical composition and evolutionary stage as a primary parameter for predicting fungal colonisation (section 2.3). Furthermore, in describing the behaviour of MBC from this perspective, we review studies having shown clear

correlation between the mycelium mechanical properties and their cultivation substrates chemical composition (section 2.4). While these properties have been demonstrated to be modifiable to result in foam-like or elastomer-like behaviour, the contribution of substrate aggregates nature, size, distribution, and shape are the principal contributors of strength and stiffness (section 2.5). Outstanding mechanical results from the MBC state-of-the-art are then discussed with regards to these insights, in an effort to describe the composite behaviour. We identify three main substrate-based exploration strategies for the MBC system: densification (by dense packing, cold or hot-pressing), composition (by introducing structuring elements), and supplementation (targeting mycelium properties, and based on chemical tuning of the substrate)(section 2.6). We are reporting on the experimental investigation of the effect of composition over the mechanical behaviour in compression and flexion (sections 5.1 and 5.3). Mogu contributed to this report by supplying a series of specimens for the flexural characterisation series. Because the effect of substrate particles size has not yet been investigated, we tested its impact in compression and tension (sections 5.1 and 5.4). With a standard protocol, we tested a change in strains on the best performing granulate size (section 5.2). Similarly, and as a way to verify the impact of the nature of lignins on white-rot fungi colonisation from a performative perspective, we tested in compression specimens of pine wood cultivated *G. lucidum* (section 5.2). In the context of the extended compression series, the principal substrate and composition materials have been qualified by Fourier-Transform Infrared (FTIR) spectrometry (section 4). The decay of *G. lucidum* on beech wood is also quantified by FTIR. Additionally, and in the continuity of the works developed in D5.1, we present a structural analysis of a kagome gridshell supporting a wet mycelium composite load. In this study we report on the effectiveness of designing hybrid kagome gridshells, made of bamboo and carbon fibre members (section 5.5). Finally, we evaluate the impact of substrate nature and aggregate geometry on thermal performances with varying thicknesses. Moreover, we report on the evaluation of two insulation buildups (section 6). Future research perspectives for the continuation of the *Fungal Architectures* project and beyond are presented in the concluding section.

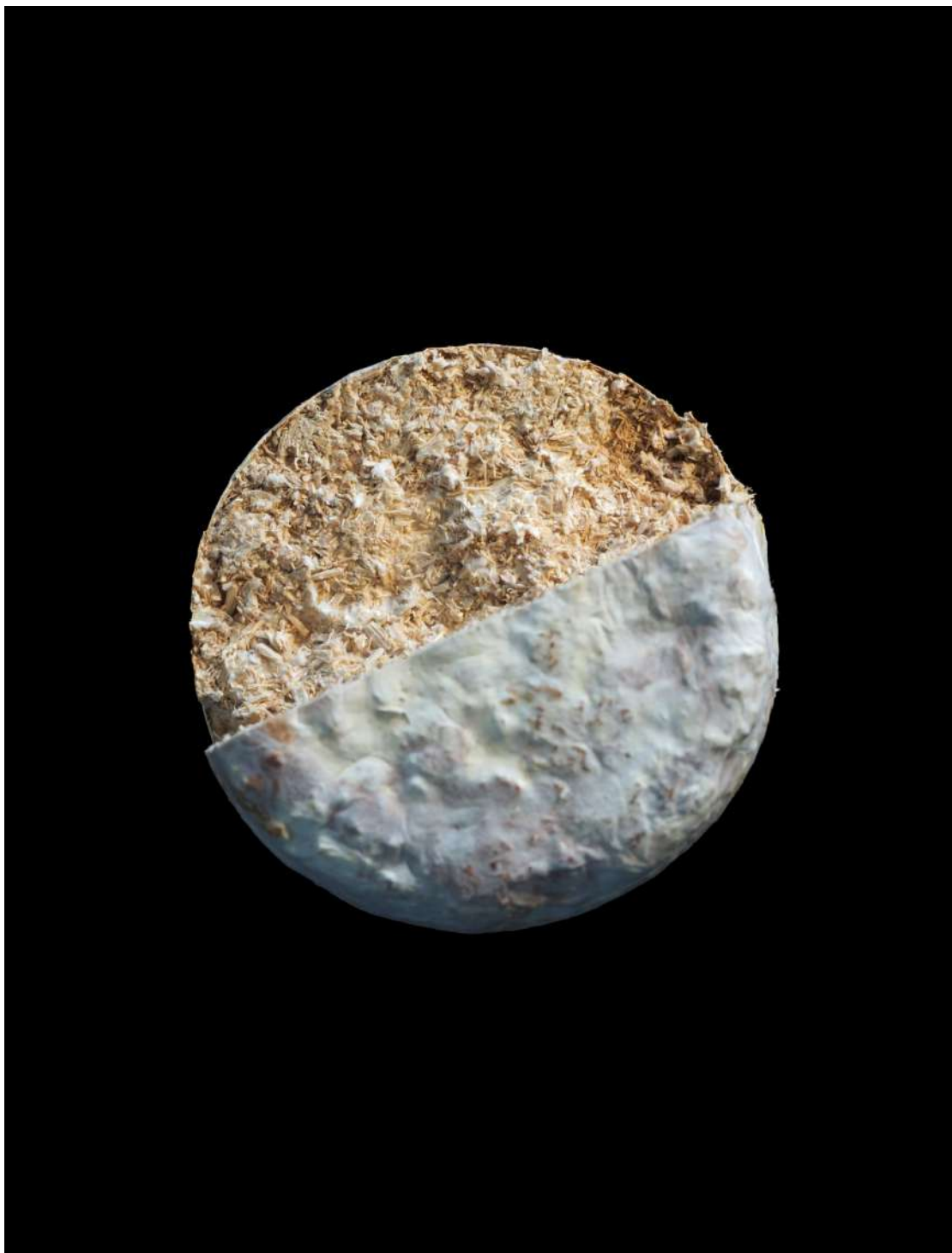


Figure 1: Typical inner structure of a MBC.

2 MYCELIUM-BASED COMPOSITE MODEL

2.1 Enzymatic activity in ligninolytic fungi

Trametes spp., *Ganoderma* spp., and *Pleurotus* spp. are among the most frequently cited families in MBC design [9]. *Schizophyllum commune* is a less investigated species but finds a growing interest [5], especially because it is one of the few mushroom-forming fungi for which genes have been inactivated by homologous recombination [87]. This strategy has proved useful for genetic studies and functional genomics [106]. Gene inactivation articulates the principles of homologous recombination for replacing an existing autologous gene with a designed heterogenous gene. Furthermore, the recombinant DNA constructs of *S. commune* has been reported to express in other mushroom-forming fungi, such as with *Pycnoporus cinnabarinus*, which supports this species importance for fungal biology [87, 2]. *Irpeex lacteus* has been used previously in MBC development [138], or even *Agaricus bisporus* [123], and *Fomes fomentarius* [84]. Other species in punctual use have been reported [9].

Wood-rot fungi can be classified by their decay activity, which cover three main modes: white-rot (specified as selective delignification, and simultaneous rot), brown-rot, and soft-rot. Across the surveyed species, all of them are white-rot, ligninolytic fungi. They have the enzymatic ability to decompose all three principal polymer classes found in wood: lignin, hemicellulose, and cellulose, but most often degrade primarily lignin and hemicellulose such as reported for *G. lucidum* [26]. In such case, they qualify as selective delignifiers, as opposed to simultaneous decomposers where all three polymer classes endure similar decay rates. Brown-rot fungi uses a two-step oxidative-enzymatic mechanism for the breakdown of cellulose and hemicellulose [19]. Their lignin degradation abilities are relatively little investigated yet, and although brown-rot fungi is a polyphyletic group evolved from at least seven white-rot lineages [43], they reportedly lack more than 60 % of the genes known to be involved in white rot. Brown-rot species degrade cellulose and hemicellulose at a higher rate than white-rot ones in laboratory isolation [23]. This decay mode is most often encountered in coniferous trees. Soft-rot fungi commonly share the ability to demethylate. This mode is associated with ascomycota and asexual forms of dicaryomycota, and is distinctive for its tunnelling decay pattern through lignin layers to cellulose and hemicellulose richer areas [111]. Lignicolous fungi can benefit from a variety of enzymatic strategies to ultimately fully decompose lignocellulosic substrates. The environmental conditions for enzyme activation help define the species niche and their subsequent distribution. The principal wood-rot enzyme families are presented in Table 1.

Lignin linkages result from radical reactions, contrarily to polysaccharides polymerisation resulting from water removal. Lignin therefore does not involve hydrolytic enzymes, and cannot be used as a source of carbon or energy for most wood-rot fungi. Its decomposition serves primarily for the fungus to gain access to hemicellulose, pectin, and cellulose compounds. Lignin depolymerisation is mostly initiated by aromatic ring oxidation by peroxidases (by use of H_2O_2 or R-OOH) [62], while other aromatic compounds in the cell structure are mostly catabolised by monooxygenases and dioxygenases. Peroxidases are majoritarilly heme proteins, such as lignin peroxidase (LiP) and manganese peroxidase (MnP). LiP are known to oxidise phenolic aromatic substrates along with various nonphenolic lignin compounds, and other organic compounds with a high redox potential. MnP leads to H_2O_2 oxidising Mn^{2+} to Mn^{3+} , an oxidising compound to monomeric phenols. Mn^{3+} chelates permeate through wood cells and decay lignin selectively. These chelates cannot oxidise nonphenolic compounds, but radicals formed by Mn^{3+} oxidation can in turn oxidise benzyl alcohols and other diarylpropane structures. MnP can also peroxidise lipids [131]. Versatile peroxidase (VP), additionally to MnP and LiP, are enzymes that are Mn^{2+} specific but that can oxidise phenolic and nonphenolic compounds in its absence (as with LiP).

They are involved in oxidation of high redox potential aromatic substrates, thus reducing Mn^{2+} -independent oxidation but not affecting Mn^{2+} -dependent one [127]. Mn^{2+} -independent oxidation capabilities for phenols, small dye compounds, and amines has also been described for MnP [62], and it is predicted from sequencing that similar hybrid MnP may be present in *Trametes* spp., *Pleurotus* spp., and *A. bisporus* among other species [79, 62]. Hydrogen peroxide, necessary for LiP and MnP activities, is the product of fungal enzymatic activity. A peroxidase family with wide substrate specificity, active at pH of approximately 3, has been identified in fungi: dye-oxidising peroxidase (DyP). Additionally to oxidising peroxidase substrates, it can perform for anthraquinones. Laccases then are a very versatile family of enzymes: they are found in nearly all fungi including ones without lignin degrading abilities [12] and perform functions ranging from synthesis of melanin and other pigments, to conidia and fruiting bodies formation, and lignin decomposition indeed [36]. These belong to a multicopper enzyme family and catalyse the removal of an electron from phenolic hydroxylic groups to form phenoxy radicals which polymerise then via radical coupling. Ultimately, the reactions lead to the presence of oxidised quinones and couples oligomeric products. Laccases also have a wide substrate specificity and can act upon an extended range of phenols, diphenols, aminophenols, and aromatic compounds. When their catalytic reaction is accompanied by demethylation it may lead to ring cleavage. Tying lignin decomposition to cellulose and hemicellulose decay, cellobiose dehydrogenase (CDH) has the ability to oxidise cellobiose, cellodextrins, and lactose [139]. Hemicellulose degrading enzymes can have the ability to hydrolyse various polymers, such as β -1,4-galactosidase that degrades xyloglucan, xylan, galactomannan, and pectin [73]. More specific xyloglucanases, xylanases, mannan-degrading enzymes, and pectinases can be found in all fungi to a various extent. Typically, pectin can be hydrolysed by polygalacturonidases, but also cleaved by β -elimination by pectin and pectate lyases. Cellulose hydrolysis by fungal enzymes employs glyco-hydrolases (GH) for glycosidic bonds cleaving. Among GH families, cellobiohydrolase I (CBH I) is the major cellulase being produced in white and soft rot; it binds to crystalline cellulose and hydrolyses it into cellobiose. CBH II, found in all but brown-rot fungi, binds to amorphous cellulose and separates cellobiose from the non-reducing end of cellulose using acid catalysis of the β -1,4-glycosidic bond. Endoglucanases are a major enzyme group able to hydrolyse cellulose polymer from within the molecule rather than its end. The variety encountered in fungi refers to four GH families: GH5, GH7, GH12, and GH45. GH5 endoglucanase is a less studied family as per fungi. Because it belongs to the same GH family, GH7 endoglucanase displays a similar mechanism to CBH I, with the difference that it can hydrolyse from the middle of cellulose molecules thanks to its structure. GH12 endoglucanase lacks a carbohydrate-binding module (CBM), they are therefore unable to degrade crystalline cellulose but can hydrolyse it in its amorphous form. GH45 endoglucanase benefits from a CBM. β -glucosidases, produced by all fungi, belong to GH1 and GH3 families. GH1 β -glucosidases are able to cleave soluble β -linked oligosaccharides from up to nine glucose residues chains and aglycone-linked β -glucosides, and can be competitively inhibited by the products of CDH activity. GH3 β -glucosidases can remove single glucosyl residues from the non-reducing ends of oligo- and polysaccharides such as β -D-glucans, β -1,3-D-glucans, β -1,4-D-glucans, to name a few [64]. Lytic polysaccharide monooxygenases (LPMO) have been shown to enhance cellulose degradation [57, 28].

Carbohydrate-active enzymes (CAZymes) have been classified in six principal families: Auxiliary Activities (AA), Glycoside Hydrolase (GH), Glycosyltransferase (GT), Polysaccharide Lyase (PL), Carbohydrate Esterase (CE), and Carbohydrate-Binding Module (CBM). We summarised the genome distribution of *T. versicolor*, *P. ostreatus* PC15, and *G. lucidum* G0119 according to this nomenclature in Table 2. The *G. lucidum* species benefits from the widest known CAZymes array with 565 genes having been assigned to these functions [141, 26]. This species is a well known versatile fungus capable of selective delignification – we confirm this experimentally in

section 4. The actualisation of decay strategies in selective delignification capable fungi has been correlated to wood species, lignin contents, temperature, pH, and moisture content [44, 92, 112, 1, 15].

2.2 Lignocellulosic substrates

Lignin is the most effective barrier to fungal decay as it prevents migration of low molecular weight diffusible agents for decomposing cellulose and hemicellulose. The nature of lignins in presence is of considerable influence on their sensitivity to enzymatic activity. Lignins are polymers of phenylpropene units, they comprise: guaiacyl, syringyl, and p-hydroxyphenyl units. Gymnospermous woods (conifers) are almost exclusively composed of guaiacyl monomers, while angiospermous trees (hardwood) contain approximately equal shares of guaiacyl and syringyl monomers [132]. Similarly, different distributions of hemicellulose compounds are found in gymnospermous trees (where galactoglucomannan, glucomannan, and arabinoglucuronoxylan are the principal ones) and in angiospermous trees (where arabinogalactan, xyloglucan, and various glucans are the principal compounds) [111]. Pectins can be accounted for too, and are primarily made of β -1,4-D-galacturonase acid units and their methyl esters, interrupted locally by 1,2-linked L-rhamnose units [53]. And the primary constituent of wood is cellulose, constituted by repeating glucose units joined by hydroxyl linkages. Cellulose represents about 50 % of the wood cell wall, lignin 25 %, hemicellulose 20 to 25 %, and pectin 1 to 4 %. Broad-leaf plants such as hemp are often in use in MBC practice. Their tow lignocellulosic compounds ratio C:HC:L is close to 15:2:1, when their shives display a ratio closer to 2:1:1. The cultivation period and weather has been reported to induce differences in non-cellulosic composition [129].

The variety of wood structures and chemical compositions pairs with various decay strategies driven from the set of fungi CAZymes expression. These wood specifics can be understood from their evolutionary history and subsequent specialised cell contents. Principle wood structures are presented in Fig.3. Wood cells can be made of various specialised cells: tracheids, xylem ray parenchyma, axial parenchyma, and vessels. Tracheids are found down to the first evolutionary stage of trees in gymnospermous woods, where they perform both water transport and structural stiffness. Gymnospermous woods typically lack vessels and rarely have axial parenchyma, their tracheids to xylem ray parenchyma ratio is most often in the vicinity of 9:1. They contain principally guaiacyl lignin, a product of coniferyl alcohol [40]. In more recent angiospermous woods, water is transported by vessels while tracheids perform structurally. The second evolutionary stage of water transport system in trees can be exemplified by European beech (*Fagus sylvatica*) and birch (*Betula pendula*). Their xylem includes vessels along with tracheids, the former being the only distinct location of guaiacyl lignin. Tracheids and parenchyma contain a fraction of syringyl lignin spread across the cell wall layers, making this secondary evolutionary wood structure attractive for white-rot fungi [111]. The third evolutionary stage can be illustrated by European oak (*Quercus robur*), for which the xylem is ring-porous containing early and late wood vessels in bands within a thin-walled tracheids matrix. The strength of the wood comes from libriform fibres containing higher syringyl lignin, while tracheids have a higher guaiacyl content [108]. The latest evolutionary stage is exemplified by sycamore (*Acer pseudoplatanus*) and ash (*Fraxinus* spp.), where syringyl dominates [109]. Vessels are dedicated to water transport here thanks to the presence of highly specialised cell functions, and are jacketed in living libriform fibres that are included in dead ones. Guaiacyl lignin has been reported to have a higher resistance to fungal decay [111], which has been corroborated by phylogenetic and experimental study showing that catalytic tryptophan presence in more recently evolved peroxidases (located in LiP and VP families) was more efficient at oxidising angiospermous woods while others performed better in gymnospermous ones [10]. Early stage parenchyma decomposition and a resistance

Target polymer	Enzyme family	Reaction	Substrate	Temperature	pH
Lignin	Lignin peroxidase	Oxidation	Phenol, nonphenolic aromatics.	28 – 40 °C	3.0 – 4.5
	Manganese peroxidase	Oxidation	Manganese ions, phenol, small dye compounds, amines.	23 – 40 °C	5.0 – 7.0
	Versatile peroxidase	Oxidation	Manganese ions, phenol, nonphenolic aromatics.	35 °C	7.0
	Dye-oxidising peroxidase	Oxidation	High redox-potential dyes, nonphenolic lignin model compounds.	28 – 70 °C	3.0 – 7.0
	Laccase	Oxidation	Phenol, diphenol, aminophenol, aromatic compounds.	35 – 50 °C	4.0 – 5.0
Hemicellulose	Xyloglucanase	Hydrolysis	β -1,3-glucans, carboxymethyl cellulose, xyloglucan oligosaccharides.	37 – 75 °C	3.0 – 9.4
	Xylanase	Hydrolysis	Glycosidic bonds in xylan backbone.	28 – 50 °C	3.0 – 9.0
	Mannan-degrading	Hydrolysis	Mannan, galactomannan.	45 – 92 °C	2.4 – 7.5
	Polygalacturonidase	Hydrolysis	Pectin.	–	–
	Pectin lyase	β -elimination	Heavily methyl-esterified.	50 °C	8.0
	Pectate lyase	β -elimination	Lower esterification and Ca^{2+} .	30 – 90 °C	5.0 – 11.5
	Cellobiose dehydrogenase	Oxidation	Cellobiose, cellodextrins, lactose.	60 °C	3.0 – 10.0
Cellulose	Cellobiohydrolase I	Hydrolysis	Crystalline cellulose.	40 – 80 °C	–
	Cellobiohydrolase II	Hydrolysis	Amorphous cellulose, β -xylan.	50 – 63 °C	4.0 – 6.0
	GH5 endoglucanase	Hydrolysis	Cellulose.	45 – 65 °C	4.5 – 7.0
	GH7 endoglucanase	Hydrolysis	Crystalline cellulose.	45 – 65 °C	4.5 – 7.0
	GH12 endoglucanase	Hydrolysis	Amorphous cellulose.	45 – 65 °C	4.5 – 7.0
	GH45 endoglucanase	Hydrolysis	Crystalline cellulose.	45 – 65 °C	7.0
	GH1 β -glucosidase	Hydrolysis	β -linked oligosaccharides, aglycone-linked β -glucosides.	25 – 50 °C	5.0 – 7.0
	GH3 β -glucosidase	Hydrolysis	β -glucan.	25 – 50 °C	5.0 – 7.0

Table 1: Extracellular enzymes in lignicolous fungi and their optimal activation temperature and medium pH (adapted from [73, 139, 127, 49, 91, 130, 71, 98, 74, 94, 30, 96, 85, 83, 137, 135, 22, 86, 59, 124]).

CAZy family	Enzymes	TV	PO	GL
AA	Laccase, LiP, MnP, VP, DyP, CDH, LPMO.	89	114	96
CBM	GH5-GH7-GH45 endoglucanase, CBH I	47	82	33
GH	GH12 endoglucanase, β -glucosidase, xyloglucanase, xylanase, polygalacturonidase, mannan-degrading, CBH II.	222	235	262
GT	Glycosyltransferases.	85	67	72
PL	Pectin lyase, pectate lyase.	9	23	11
CE	Carbohydrate esterases.	19	28	91

Table 2: Genome distribution of CAZymes from selected species: *T. versicolor* (TV) [117], *P. ostreatus* PC15 (PO) [117], *G. lucidum* G0119 (GL) [141].

from libriform fibres to white-rot decay has been confirmed experimentally [111]. The structure of the wood cell wall composing the xylem divides between the middle lamella (ML), primary wall (P), and three layers of secondary wall (S1, S2, S3). They are represented with their fibre orientation and indicative thicknesses in Fig.2. The structuring middle lamella is composed of lignin, calcium, and pectic compounds that are amorphous. Crystalline cellulose is most prominent in younger wood, so it dominates S3, and is found in lowering percentages moving towards the P. Inversely, hemicellulose can be found in greater extend in the P and in reducing shares towards S3, while lignin is quite evenly spread across the cell wall, with higher concentrations in the ML. The distribution of lignocellulosic compounds in woods is presented in Table 3, and an illustration of wood cell wall specification in arboreal plants is presented in Fig.4. Colonisation happens through pit apertures present in wood, or the inducing of boreholes by enzymatic activity. Although pectin contents are marginal in wood, pectinases expression is critical to decomposing the pit membrane (Fig.2), which also contains cellulose, and allow the fungus to proceed in colonising [113]. Calcium is associated with pectin in wood, which can be hydrolysed by fungal oxalic acid, leading to Ca^{2+} chelates [53]. The main plant fibre minerals include calcium, potassium, phosphors, and magnesium. The mineral content (ash) in softwood is 0.02 – 1 wt% and 0 – 5 wt% in hardwood [128]. We can note that large concentrations of calcium oxalate and black residues of MnO_2 accumulate during white-rot decay [15]. Wood can moreover contain a variety of toxic compounds, mainly polyphenols or tannins in angiospermous trees and phenolic compounds in gymnospermous ones. Selective delignification capable fungi can degrade polyphenols thanks to their chemical proximity to lignin. *Ganoderma* spp. are remarkable not only for having this ability, but also because they have been reported to be attracted to these compounds [110]. Nitrogen being necessary to fungi for enzyme synthesis, the limited amounts in wood makes it resilient to decay (C:N of birch is typically of 55, and sycamore 401). This compound is mostly found in parenchyma and is known to migrate towards the bark surface during log desiccation [16]. To fill in for a lack of available nitrogen, white-rot fungi such as *P. ostreatus* have been shown to be able to trap and feed from nematodes, while the latter are usually ones to feed from their mycelia [13].

As previously suggested, the wide distribution of wood-decaying fungi suggests with increasing evidences that environmental factors and substrate specifics drive and regulate fungal colonisation between various fungal species in presence. Furthermore, while the pH of wood is most often in the 4.0 – 5.5 range (beech: 5.11, birch: 5.29, for instance) [46], it has been demonstrated that substrate pH is also influenced by the action of decay [44]. It has been reported by comparative transcriptomics that white-rot related genes of a *Phanerochaete carnosa* species would all express in fir (*Abies balsamea*), pine (*Pinus contorta*), spruce (*Picea glauca*), and maple (*Acer*

saccharum) media, but to a different extent. For instance, MnP genes read was 2.42 times higher in fir compared to maple within the 30 most expressed genes, but LiP related genes read was 4.7 folds higher in maple [80]. This indicates both a resilience of species to the use of a variety of vernacular substrates, but also a precariousness considering the sheer distribution of competitors of an individual species. In its most extreme cases this leads to intense fungus chemical wrestling for resource [61], a battle in which *Ganoderma* spp. has been recognised the advantage of being able to degrade reaction zones [110]. Studies of lignicolous fungi ecological distribution point to water content and temperature variations being the principal drivers of selection [17]. Climate change has then been related to shifts in fungal niches across hosts [18]. In the wild, host tree species and their evolutionary stage can have an influence on patterns of decay: the latest evolutionary stage that has the most diffuse xylem vessel system may result in a slower initial colonisation. Similarly, gymnospermous wood having 3 – 4 mm long tracheids it benefits from a cellular compartmentalisation strategy to oppose fungal decay, additionally to its guaiacyl lignin content. These topological effects are prominent at the early stage of colonisation [16].

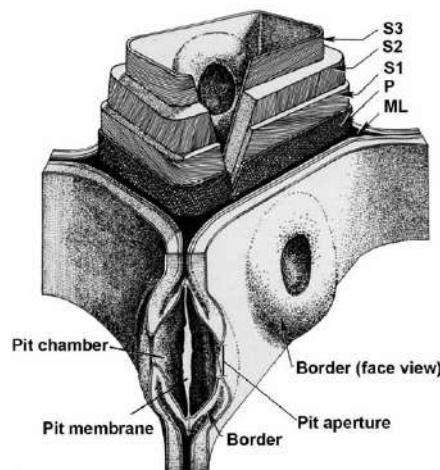


Figure 2: Wood cell wall structure: middle lamella (ML), primary wall (P), secondary wall (S1, S2, S3) [133].

2.3 Water in fungal decay

The presence of capillary water in the substrate is a prerequisite to allow extracellular transport of metabolites in fungi, and to establish osmotic potential for allowing turgor pressure and subsequent hyphal growth. Turgor is also thought to enable fungi to penetrate solid substrates [82]. Moreover, the activity of water, because it allows the maintaining of turgidity of fungal cells at hyphal tip, is a critical parameter of growth. To illustrate the extent of the influence, it has been reported that the white-rot species *Physisporinus vitreus* radial growth at pH 5 and 25 °C increased from just below 0.5 mm/d at $a_w \leq 0.974$, to 4.5 mm/d at $a_w = 0.998$ [45]. Furthermore, the production of enzymes has been related to water activity [47].

SSF requiring the absence of solution in between particles for a mycelium to develop, the substrate water holding capacity is principal parameter for ensuring an optimal fungal colonisation and predicting fungal growth [19, 39, 47]. In timber, moisture can exist as bound water within cell walls below fibre saturation point (FSP), free liquid water in cell cavities above FSP, and

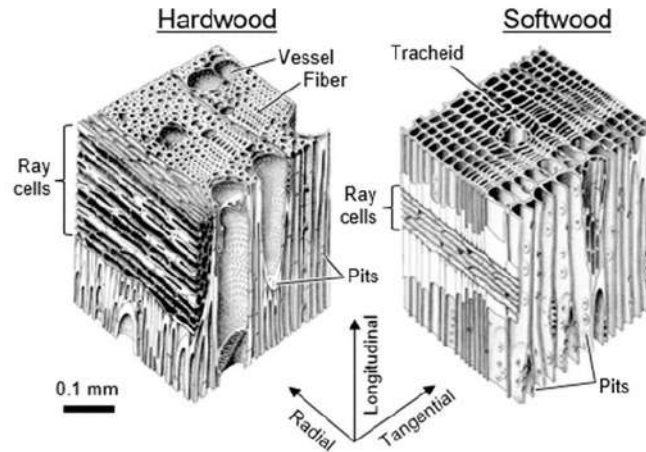


Figure 3: Principle structures of soft and hardwood. Illustration from [69].

Wood type	Cell type	Layer	Share of total lignin content (%)	Lignin concentration (%)
Gymnospermous early wood	Tracheid	S1 – S3	65	24
		ML, P	21	49
		Cell corners	14	64
Gymnospermous late wood	Tracheid	S1 – S3	75	22
		ML, P	14	51
		Cell corners	11	78
Angiospermous wood	Fibre	S1 – S2	60	19
	Fibre	ML, P	9	40
	Fibre	Cell corners	9	85
	Vessel	S1 – S3	9	25
	Vessel	ML, P	2	40
	Ray cells	S1 – S3	11	25

Table 3: Approximated lignin distribution in the wood cell wall. Adapted from [41].

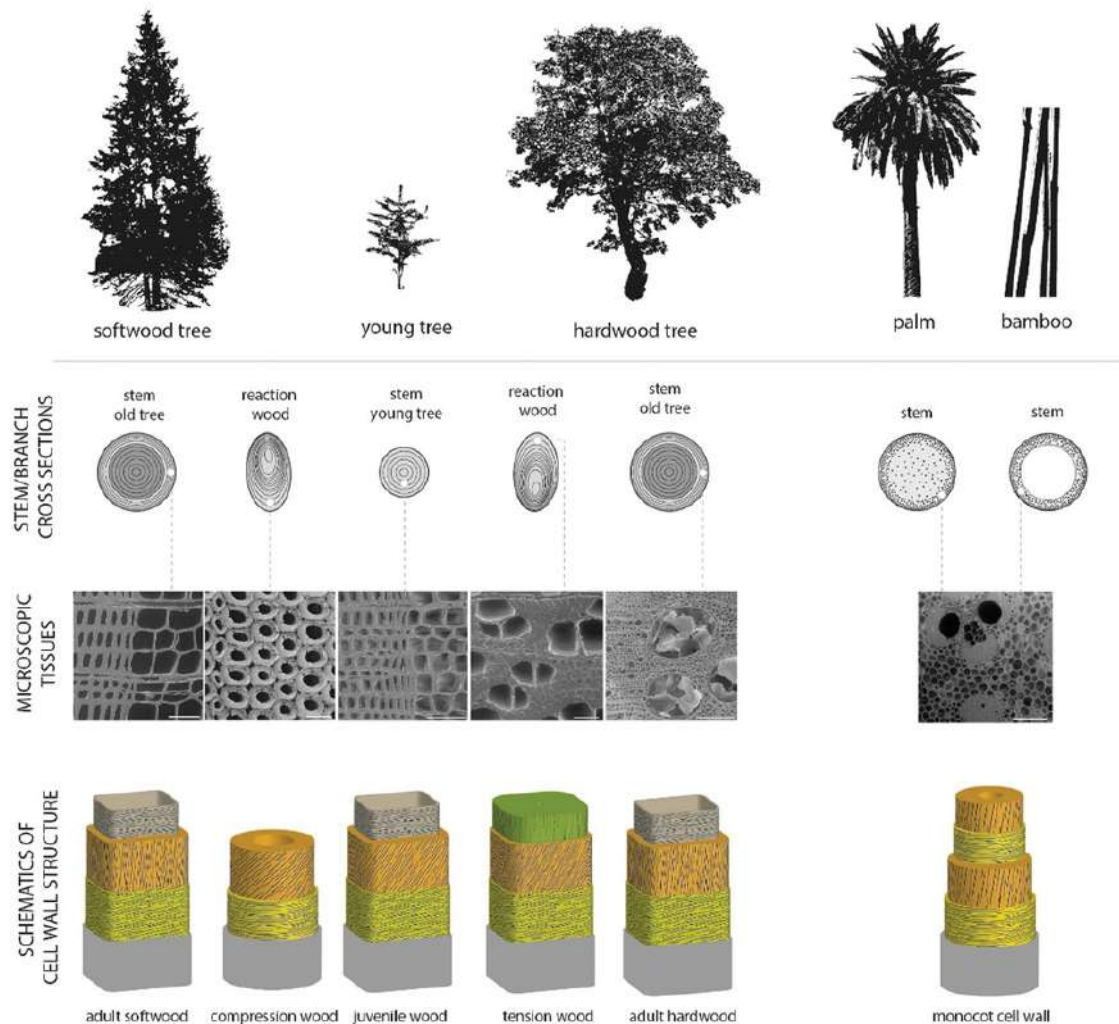
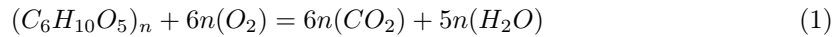


Figure 4: Structure of arboreal plants. Microscopic tissues from left to right: adult spruce wood (bar 50 μm), spruce compression wood (bar 20 μm), juvenile spruce wood (bar 50 μm), poplar tension wood (50 μm), adult oak wood (bar 200 μm), and vascular bundle of bamboo (bar 200 μm). Last row showing schematics of cell wall structures. Light gray indicates the compound middle lamella (0.5 – 1.5 μm) and primary cell wall (approximately 0.1 μm), yellow S1 layer with a thickness of 0.1 – 0.35 μm , orange S2 layer as 1 – 10 μm thick; inner light gray layer shows the 0.5 – 1.1 μm thin S3 layer; the green G-layer in tension wood fibres can fill the whole lumen and the cellulose fibrils are oriented along the longitudinal cell axis. Illustration from [35].

water vapour. Relative humidity (RH) of up to 97 – 98 % in the environment of timber is generally considered to correspond to cell wall intake (hygroscopic range), and as RH moves toward saturation, wood cell voids intake tend to become more significant in the moisture distribution (over-hygroscopic range) [126]. The latter is conventionally considered as the fittest range for optimal fungal growth in lignocellulosic SSF. The validity of this convention is nonetheless discussed as it was shown that *T. versicolor* would colonise European beech wood at a minimum moisture content (MC) of 15 % for a mass loss (ML) over 2 % [81], that brown-rot fungi grow at 92 – 97 % RH, and that instrumental limitations are encountered in the over-hygroscopic range thus complicating measures and experimental control [126]. The FSP of European beech wood has previously been determined to be 24.02 % MC at 97 % RH and 20 °C [70]. The minimum moisture threshold has been introduced as the metric having the most impact on predicting fungal growth [20, 19]. It represents the MC below which fungal decay cannot be initiated, and is typically given at a ML $\geq 2\%$.

Hydroxyl groups are the predominant sorption sites for water molecules in wood. Their amount in hemicellulose is twice as high as in lignin, and four times higher than that of cellulose [125]. It has been reported that cellulose can endure structural changes while wood is desiccated [8, 75]. It was hypothesised that lower desiccation rates would give enough time for amorphous cellulose to crystallise, rendering the substrate less hydrophilic. On the contrary, higher rates of desiccation may lead to more prominent amorphous cellulose and more hydroxyl availability. Lignin, hemicellulose, and pectin polymers act as a barrier to cellulose crystallisation, which reflects on the higher crystalline content towards the S3 cell wall layer [104]. Because hemicellulose have a softening point at room temperature and 18 – 20 % MC [39], while lignin softens above 60 °C when saturated by water, and cellulose is hardly softened in its crystalline configuration (≥ 200 °C), higher thermal treatments may result in hemicellulose degradation thus rendering the substrate less hydrophilic [95].

Finally, the amount and quality of water in the SSF system varies during fermentation. We can note that fungal decay by oxidation produces water along with CO₂:



C₆H₁₀O₅ being the repeating unit of glucose polymers, such as found in starch or cellulose. As bound water is freed too during decay, and while substrate dry ML is occurring, the MC is increasing if the system is closed. The quality and location of water in SSF systems are then a prime set of information to predict and then foster or prevent fungal development. They are of primary interest to design and produce MBC, while to the best of our knowledge no study investigating the role of water in MBC production has been reported yet. The optimal moisture content for laccase, LiP, and MnP activation is consistently reported to be 60 – 75 % [56, 66, 65, 7].

2.4 Mycelium mechanical properties

It is clear that mycelium acts as a binder between wood particles: it does so by degrading lignocellulosic compounds and producing a dense hyphal network that interlocks dispersed particles. The binding between two particles is not distinctly strong at the unit scale, but the very high redundancy in the composite can result in a mechanically valuable material. It has been reported that a *P. ostreatus* mycelium would be more than two times stiffer than a *G. lucidum* one when both cultivated on a cellulose medium [55]. Authors comment this difference with regards to the higher protein and lipids content in *G. lucidum* that can act as plasticisers. Furthermore, in this study a set of experiments grown on a PDB-cellulose medium has been compared to the previously mentioned results, and it was found to decrease the Young's modulus of the *G.*

Tensile behaviour	PO*	PO**	GL*	GL**	ND†
Young's modulus (MPa)	28	17	12	4	0.6 – 2.0
Ultimate strength (MPa)	0.7	1.1	1.1	0.8	0.1 – 0.3
Elongation at brake (%)	4	9	14	33	–

Table 4: Tensile behaviour of mycelia from *P. ostreatus* (PO) [55], *G. lucidum* (GL) [55], and a non-disclosed species (ND) [67]. *Cellulose medium. **Cellulose and PDB medium. †Non-disclosed medium.

lucidum mycelium film by a factor of 3 while increasing its elongation at brake by 2.3 folds. This corroborates their previous comment on plasticisers, and suggests that nutrition medium tuning can help specify composites behaviour further than solely seeking stiffness. This study suggests that the matrix phase of the composite can be made closer to an elastomer or stiffened by varying protein, lipids, or alcohol contents. Similar findings have been reported [3]. Isolated mycelium mechanical behaviour has been investigated so as to model it [68]. At the resolution of hyphae, researchers have identified three main stages of response of a ligninolytic fungi mycelium: linear elastic behaviour at small strains (hyphae bending), fibre buckling and local structural collapse or densification at larger strains, and rapid stiffening associated with full compaction and large number of inter-fibre contacts. This model reflects upon the open-cell foam model. To fit their FEA simulation model, an hyperelastic model was used in this study as a mycelium undergoes large displacements in its elastic behaviour and results in a non-linear stress-strain response. This study relies on the characterisation in tension and compression of an aerial mycelium. Young's modulus situate in the 0.6 – 2.0 MPa range for a density in 30 – 50 kg/m³, the yield stress in 40 – 80 kPa, and ultimate tensile strength in 100 – 300 kPa [67]. The results are presented in Table 4.

2.5 Substrate mechanical properties

Most commercially interesting woods have a Young's modulus situated in 5.5 – 15.7 GPa at 12 % MC, and 4.4 – 12.3 GPa when green [52]. The distribution of these performances is similar in gymnospermous and angiospermous woods. Wood particles are therefore important load-carrying members of the composite system and reduce the magnitude of stress experienced by the mycelial matrix. The plastic strain of the composite is contributed to only by particles in such composite. The dewetting behaviour of the larger particles present in the composite is a principal contributor to damage nucleation in such two-phase particulate composite. Furthermore, the shape, nature, and distribution of particles in two-phase particulate composites has been shown to have a substantial influence over the load transfer between members and hence their overall stiffness [115]. Moreover, while lignin is a primary contributor of strength parallel to grain, hemicellulose supports compression strength perpendicular to grain. Its decay affects greatly the structural integrity of wood and its hardness [29].

In particle studies of granular materials, shape parameters such as flakiness/flatness (ratio of particle thickness to width), elongation (length to width ratio), sphericity (deviation from a sphere geometry), and roundness/angularity (measure of angles sharpness) have been investigated [114]. Without matrix phase, studies have reported that a 3:1 ratio of flaky particles content would be an approximate optimal for shear strength (depending on the system of study). This was related to cohesion increase under stress related to particles interlocking. Furthermore, it was reported that a higher particle angularity leads to a decrease in elastic modulus, while ultimate strength increases. The shear strength was reported to follow the same dynamic, increasing

with particle angularity. Increasing flakiness and angularity leading to increased cohesion, it is also subjected to a higher abrasion and leads to damage accumulation under repeated loads, thus leading to strain accumulation [114].

While wood particles are of common use in MBC, other more elastic substrates have been used such as non woven cotton fibres [4]. In commercially available hemp shives too, flexible residues of hemp tow processing can be found. In such cases, the particle may not contribute structurally and the resulting composite behaviour will be driven by the mycelium and its chemical expression, they nonetheless contribute to the composite strength in tension.

Few researches have also been investigating the addition of non-organic aggregates to a lignocellulosic substrate for improving its stiffness, such as with carbonate sand [50] (which also acts as a pH buffer being mainly composed of CaCO_3), and sand and gravel [84]. It was reported that supplementation of a cotton and wheat bran medium with 37.5 wt% carbonate sand resulted in a 1.6 fold stiffening, and a factor 4 increase in ultimate compressive strength, while density increased by 27 %.

2.6 Composite mechanical properties

The large number of influential variables on the mechanical properties leads to the emergence of a variety of methods for tuning them. While common practice leads to a maximum of compressive modulus of maximum 2 MPa, post-cultivation heat-pressing has been reported to lead to Young's modulus of 35 – 97 MPa and flexural modulus of 34 – 80 MPa [4]. Use of particles-additives strategy additionally to a wheat bran supplemented medium made of cotton stalk has proved significant: an elastic modulus of 48.5 MPa was reported with 37.5 % carbonate sand addition with *P. ostreatus*, while a non supplemented control in this study reached 30.3 MPa [50]. A second study reported on an elastic modulus situated in the 39 – 60 MPa range for a density in 240 – 265 kg/m³ [138]. Authors of the latter are elusive about production details, but nonetheless indicates the use of Alaska birch shavings (*Betula neoalaskana*), millet grain, wheat bran, calcium sulphate, and the addition of a natural fibre. The specifics of the performative batch are not disclosed, but a pre-colonisation was performed prior to moulding. The performance does not come from a single factor considering the experimental plan. The considerable use of fibres (of undisclosed quality), grains, brans, and CaSO_4 , has us understand this as an hybrid improvement using both substrate based reinforcement with the fibres, and additives. Other stiffening strategies have been investigated, such as with the introduction of SBR Latex, a bonding agent for the construction industry. With a 5 % addition of SBR Latex and 0.5 % silane coupling agent to a cotton seed hull medium, the cultivation of a *P. ostreatus* species displayed a 2.39 factor increase in strength with a 27.6 % density increase [58]. The resulting environmental impact was not discussed in the study. As another high impact material [6], cellulose nanofibrils (CNF) addition has been reported [121]. Between two specimen groups, the optimal supplementation of CNF was reportedly of 2.5 – 5 wt% with a contribution in Young's modulus of a factor 2.56 – 5.6 and 2.8 – 7 in ultimate strength. In this study, the effect of densification on composites embedding a 2.5 wt% CNF part was reported: densifying from 300 kg/m³ to 600 kg/m³ contributed to increase exponentially the strength to a factor 17, while the stiffness increase by 14.38 folds. The elastic modulus in tension of MBC is reportedly situated in 3.0 – 13.0 MPa, these values resulting from a densified 24 days cultivated *T. versicolor* mycelium on wheat bran supplemented beech sawdust [4]. For this specimen group, a higher flexural modulus of 9.0 MPa was reported. The state-of-the-art in MBC research largely considers monolithic and homogeneous composites, besides a few study groups investigating jute type materials in sandwich composite reinforcement, and wood panels introduction [72, 103, 142].

Across the state-of-the-art we can identify three main research strategies: densification (by com-

posite packing, cold or hot-pressing), composition (by introducing structuring elements such as fibres and aggregates), and supplementation (chemical tuning of the substrate). The contribution of these strategies on the compressive elastic modulus are illustrated in Fig. 5. These are substrate focused strategies, targeting mycelium properties tuning in the case of supplementation, while the very choice of the fungal species can result in radically different cultivation lead time and mechanical properties too. Genetic modification of fungi could promote specific enzymatic activity or modify the composition of its cell wall to help improve performances, while the investment cost and ethics of this practice may lead to prioritising more frugal and affordable techniques.

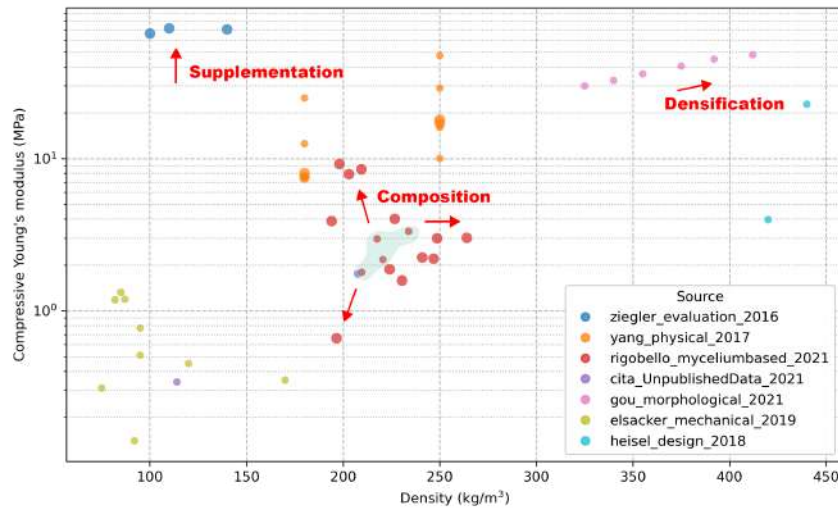


Figure 5: Effect of exploration strategies on the compressive Young's modulus. Data retrieved from [142, 138, 100, 50, 38, 60]; *cita_UnpublishedData_2021* refers to values currently only reported in this document. Larger points correspond to structured composites (such as textile or fibre reinforced).

3 MATERIALS AND METHODS

Across the three strategies (densification, composition, and supplementation), we are investigating the effect of composition over the mechanical behaviour in compression and flexion (sections 3.5 and 3.6). Because the effect of substrate particles size has not yet been investigated, we tested its impact in compression and tension (sections 3.5 and 3.7). With a standard protocol, we tested a change in strains on the best performing granulate size (section 3.5). Similarly, and as a way to verify the impact of the nature of lignins on white-rot fungi colonisation from a performative perspective, we tested in compression specimens of pine wood (*Pinus sylvestris*) cultivated *G. lucidum* (section 3.5). In the context of the extended compression series, the principal substrate and composition materials have been qualified by Fourier-Transform Infrared (FTIR) spectroscopy (section 3.4). The decay of *G. lucidum* on beech wood is also quantified by FTIR.

A variety of experimental designs are being used in the field of MBC research to evaluate mechanical characteristics. Few studies consider material evaluation standards, and systematically derive from the standards guidelines (regarding specimen or experimental design). The referenced standards in the state-of-the-art are presented in Table 5. Among the three strategies introduced previously, we have decided to investigate both the effect of composition strategies over the behaviour in compression and flexion, and the effect of aggregate size over compression, tension, and flexion. The ASTM that is the fittest for evaluating these properties with adopting the two-phase particulate composite model is ASTM D1037. It has the benefit of being the most referred set of guidelines in MBC development and covers various tests. The experimental plan is presented in Fig.6, and its colour legend is reported in Table 6.

In addition to the investigation of mechanical properties of MBC, we have conducted studies on thermal performance of MBC with a view towards building insulation application. We extend the scope of the studies reported in the literature (see Table 7), which focus on material properties, to consider hypothetical building envelopes composed of MBC. This effort will directly inform design approaches being developed as part of T5.3 - Design Rules for Fungal Architecture. As such, we follow ISO 9869 for determining thermal resistance and thermal transmittance via in-situ measurement. We design a two stage experimental plan that determines thermal properties of two material series, and then tests two configurations of a hypothetical building envelope build-up (section 3.9).

3.1 CITA cultivation method: mechanical tests

Materials

Millet-grown spawns of species *G. lucidum* (reference M9726) and *G. resinaceum* (reference M9732) were acquired from Mycelia BVBA (Nevele, Belgium). Spawns were stored at a constant 4 °C and 65 % RH. The principal substrate of the specimens is European beech wood (*Fagus sylvatica*). We used three granulations (small, medium, large): 0.5 – 1.0 mm (Räuchergold type HB 500/1000, J. Rettenmaier & Söhne GmbH + Co KG, Rosenberg, Germany), 0.75 – 3.0 mm (Räuchergold type HB 750/2000, J. Rettenmaier & Söhne GmbH + Co KG, Rosenberg, Germany), and 4.0 – 12.0 mm (Räuchergold type KL 2/16, J. Rettenmaier & Söhne GmbH + Co KG, Rosenberg, Germany). A fourth particle type was added to the experimental plan, as a 1:1:1 volume ratio mix of the three granulations. Longitudinal fibres were introduced in a specimen series by using common reed fibres (*Phragmites australis*; Tækkemand Chresten Finn Guld, Køge, Denmark). 6 mm diameter rattan fibres have been used for compression and flexion testing (*Calamus manan*; B.V. INAPO, Bloemendaal, Netherland), such as hemp-based hessian (*Cannabis sativa* subsp. *sativa*; NEMO Hemp jam web 370 g/m², Naturellement Chanvre,

Test	Standard	Designation	Ref.
Compression	ASTM D3501	Standard Test Methods for Wood-Based Structural Panels in Compression.	[38]
	ASTM D695	Standard Test Method for Compressive Properties of Rigid Plastics.	[119]
	ASTM C67	Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile.	[88]
	ASTM D2166	Standard Test Method for Unconfined Compressive Strength of Cohesive Soil.	[138]
	ASTM D1037	Standard Test Methods for Evaluating Properties of Wood-Base Fiber and Particle Panel Materials.	[142, 25]
	ASTM C165	Standard Test Method for Measuring Compressive Properties for Thermal Insulations.	[63]
	EN 826	Thermal insulating products for building applications – Determination of compression behavior	[37]
	ASTM D3574	Standard Test Methods for Flexible Cellular Materials—Slab, Bonded, and Molded Urethane Foams	[60]
Flexion	ASTM C203	Standard Test Methods for Breaking Load and Flexural Properties of Block-Type Thermal Insulation.	[63]
	ASTM C393	Standard Test Method for Core Shear Properties of Sandwich Constructions by Beam Flexure	[72]
	ASTM D7250	Standard Practice for Determining Sandwich Beam Flexural and Shear Stiffness	[72]
	ASTM D1037	Standard Test Methods for Evaluating Properties of Wood-Base Fiber and Particle Panel Materials.	[77, 122, 25]
	ISO 16978	Wood-based panels – Determination of modulus of elasticity in bending and of bending strength.	[37]
	ISO 12344	Thermal insulating products for building applications – Determination of bending behavior.	[37]
Tension	ASTM D1037	Standard Test Methods for Evaluating Properties of Wood-Base Fiber and Particle Panel Materials.	[142, 25]

Table 5: Referenced standards for mechanical property determination in the MBC state-of-the-art.

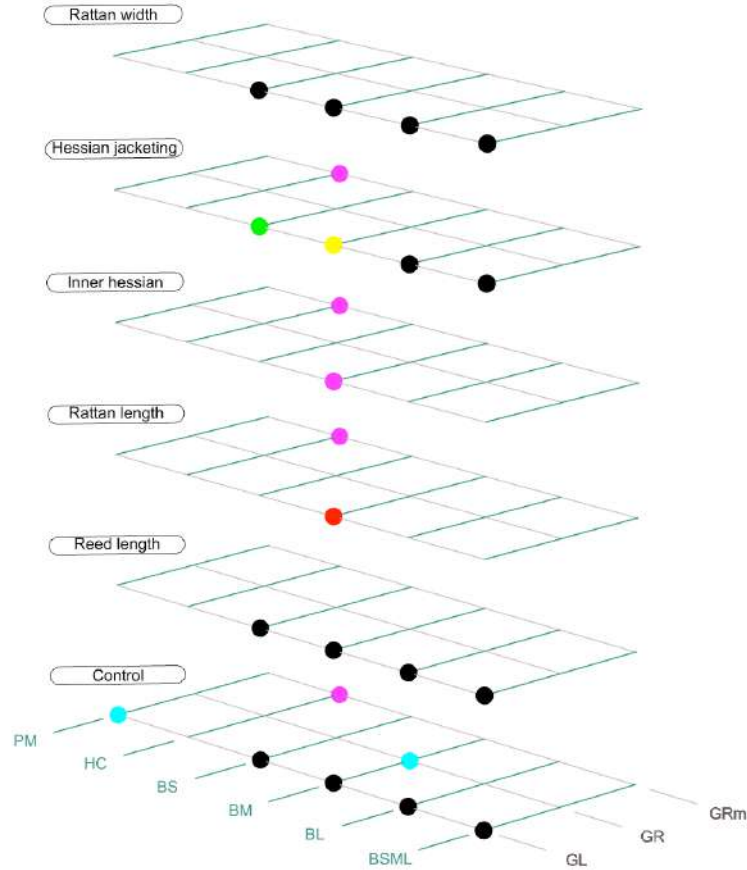


Figure 6: Distribution of experimental characterisation series across fungal species (GL: *G. lucidum*, GR: *G. resinaceum*, GR_m: *G. resinaceum* from Mogu records), substrate types (PM: pine shavings, HC: hemp fibres supplemented with coffee ground, BS: beech wood small, BM: beech wood medium, BL: beech wood large, BSML: 1:1:1 volume ratio mix of BS, BM, and BL), and fibre composition strategies (planes). The colours legend is reported in Table 6.

Colour	Flexion	Compression	Tension
Magenta	X		
Black		X	
Cyan			X
Red	X	X	
Green		X	X
Yellow	X	X	X

Table 6: Experimental plan test types correspondence table.

Test	Standard	Designation	Ref.
Thermal Conductivity	ASTM D5334-14	Standard Test Method for Determination of Thermal Conductivity of Soil and Soft Rock by Thermal Needle Probe Procedure.	[138]
	ASTM D5334-00	Standard Test Method for Determination of Thermal Conductivity of Soil and Soft Rock by Thermal Needle Probe Procedure.	[38]
	ISO 8302/EN 1946-3	Thermal insulation — Determination of steady-state thermal resistance and related properties — Guarded hot plate apparatus.	[31]
	DIN EN 12667	Thermal performance of building materials and products - Determination of thermal resistance by means of guarded hot plate and heat flow meter methods - Products of high and medium thermal resistance.	[102]
	ASTM C518-17	Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus.	[134]
	None referenced	Study conducted using EP500 hot plate apparatus (Lambda-Messtechnik).	[97]
	None referenced	Study conducted using KD-2 Pro thermal analyser.	[136]
	None referenced	Study conducted using TPS 500 thermal conductivity instrument.	[63]
	None referenced	Study conducted using dynamic hot-wire method.	[107]
Specific Heat Capacity	None referenced	Study conducted using KD-2 Pro thermal analyser.	[136]

Table 7: Overview of literature covering thermal property determination of MBC for thermal insulation applications, together with referenced standards where declared.

Echandelys, France). The materials used by CITA for the mechanical and thermal studies are illustrated in Fig.7.



Figure 7: Materials used by CITA for the mechanical and thermal studies of this report (left to right, top to bottom): beech large (BL), beech medium (BM), beech small (BS), birch shavings, hemp shives, rattan fibres, hessian, and common reed fibres.

Protocol

The principal substrates, fibres, and hessian were prepared at 70 % MC with mineralized water, and sterilised at 121 °C for 15 min. The principal substrates were then mixed with 16 wt% spawn and incubated in polypropylene filtered bags (SacO2, Deinze, Belgium) for 7 days at 25 °C in the dark. Once colonised, the principal substrates were broken down and formed with the sterile fibre and hessian into alcohol cleaned aerated PETG moulds. The formed specimens were incubated for 21 days at 25 °C in the dark, then oven dried for 48 hours at 60 °C. The dried specimens were stored at 4 °C and 65 % RH prior to testing. No external mycelium was cultivated on the external boundaries of the specimens. No additive was used. The same cultivation protocol was used for all mechanical characterisation test related specimens (compression, flexion, tension).

3.2 CITA cultivation method: thermal test

A millet-grown spawn of species *G. lucidum* (reference M9726) was acquired from Mycelia BVBA (Nevele, Belgium). The spawn was stored at a constant 4 °C and 65 % RH. Two principle substrates were selected for the specimens based on their nutritional compatibility with the selected mycelium species, and difference in dry density (see section 3.9). These were: hemp

shives (*Cannabis sativa* subsp. *sativa*) sourced from Zooplus AG, Germany; spent birch shavings (*Betula* spp.) sourced from a local distillery, Copenhagen.

Protocol

The principal substrates were prepared at 70 % MC with mineralized water, and sterilised at 121 °C for 15 min. The principal substrates were then mixed with 16 wt% spawn and incubated in polypropylene filtered bags (SacO2, Deinze, Belgium) for 7 days at 25 °C in the dark. Once colonised, the principal substrates were broken down and added to alcohol cleaned polypropylene microboxes with HEPA-filters integrated into the lids (SacO2, Deinze, Belgium). Microboxes of dimension 185x185mm at base were used, with three height variations (78 mm: model TP2000; 112 mm: model TP3000; 191 mm: model TP5000) to produce MBC panels of 40 mm, 80 mm, and 120 mm thickness respectively. The formed specimens were incubated for 21 days at 25 °C in the dark. An external mycelium was cultivated on the external boundaries of the specimens by demoulding and incubating them for another 5 days in a moisturised container. No additive was used. The specimens were oven dried for 48 hours at 60 °C. The dried specimens were stored in ambient room conditions prior to testing.

3.3 Mogu cultivation method

Materials

A *G. resinaceum* species from Mogu record 19-18 was used (GR_m). Hemp shives (cellulose: 35 %; hemicellulose: 18 %; lignin: 21 %; protein, pectin, and other products: 18 %) were acquired from HempFlax BV. (ref. 7000277; Oude Pekela, The Netherlands). Coffee ground was acquired from a local grocery store, filtered for 5 min with boiling water, and the spent coffee ground was collected.

Protocol

The principal substrates, fibres, and hessian were prepared at 60 % MC with mineralized water, and sterilised at 121 °C for 90 min. The substrates were then mixed with 3 wt% spawn and incubated for 21 days at 25 °C in the dark. Once colonised, the principal substrates were broke down and formed with the sterile fibre and hessian into aerated moulds. The formed specimens were incubated for 7 days at 25 °C in the dark, then oven dried for 72 hours at 70 °C.

3.4 Chemical analysis

Fourier-Transform Infrared (FTIR) spectrometry has been used previously for analysing the lignocellulosic profiles of substrates and their relation to fungal degradation patterns [90, 89, 55], with the benefit of requiring a limited specimen preparation, and spectra shape and frequencies being directly related to microscopical physical quantities and hence prepared for interpretation [51]. FTIR spectrometry was conducted in this study on a single reflection diamond Attenuated Total Reflectance (ATR) Agilent 4500a FTIR (Santa Clara, USA). The acquisition resolution was 4 cm⁻¹ with 16 scans per specimen, for a band between 4000 cm⁻¹ and 650 cm⁻¹. We corrected the baseline of FTIR spectra following the adaptive iteratively reweighted Penalised Least Squares (airPLS) method [140], and spectra normalisation was done with amide I/II band envelopes [11]. Four samples were isolated from *G. lucidum* colonised beech wood specimens, their spectra were averaged for analysis. The other materials were scanned with one replicate.

Chemical analysis of *G. lucidum* colonised beech was performed on specimens having been tested in compression previously.

3.5 Compression series

Following ASTM D1037, we report compression parallel to surface evaluation, for which the short-column method has been chosen as the specimens have of a nominal thickness above 25 mm. They are parallelepipeds of 1:1:4 ratio, their dry dimensions are 34 x 34 x 140 mm.

Effect of particle size & reinforcements

Four specimen groups were designed and prepared by CITA:

- Principal substrate as small granulation beech wood,
- Principal substrate as medium granulation beech wood,
- Principal substrate as large granulation beech wood,
- Principal substrate as mixed granulation beech wood,

Additionally, reinforcement strategies were investigated in four groups:

- Control: principal substrate only,
- Hessian jacketing: a hessian jacketing was introduced in the length,
- Fibres perpendicular to load: 32 mm long rattan fibres were positioned regularly within the principal substrate as two layers of fibres, centred in the specimen thickness and separated by a 10 mm layer of principal substrate,
- Fibres coaxial to load: eight to ten fibres of common reed of $1\text{ mm} \pm 0.5$ diameter were chosen so as to balance their dimensional variability and positioned in two layers separated by 10 mm of principal substrate.

No external mycelium was left to grow on the external boundaries of the specimens so as to observe the effect of experimental variables without introducing a specimen geometry bias. We identify this bias as critical for the reproducibility of experiments as the characteristics of the external mycelium mat is never found to be reported in the state of the art. A jacketing strategy has been integrated to study the effect of boundary reinforcement. Six replicates were produced and tested for each specimen type. These specimens were tested at a loading speed of 1.0 mm/min.

Effect of species

A medium granulation of beech wood cultivated *G. resinaceum* was tested with an identical cultivation protocol (BM_GR group). Pine wood sawdust was collected from a CITA milling workshop and was prepared following the previously described protocol with a *G. lucidum* species. These results are discussed with regards to *G. lucidum* grown medium beech wood (BM). 5 replicates have been tested for the BM_GR group, and only two could be tested for the PM_GL group. Load testing was performed on a Mecmesin MultiTest-dV testing bench equipped with a 2500 N load sensor. These specimens were tested at a loading speed of 4.0 mm/min. Young's modulus and ultimate compressive strength were calculated following ASTM D1037.

3.6 Flexion series

Our experimental plan investigates the effect of reinforcement strategies over the flexural modulus and ultimate flexural strength. Following ASTM D1037, we report on three points bending. To this end, four specimen groups were designed:

- Control: no fibre,
- Inner hessian: a flat layer of hessian was introduced at mid-thickness,
- Hessian jacketing: a hessian jacketing was introduced in the length,
- Rattan: five parallel rattan fibres of 6 mm diameter by 500 mm, separated by 8 mm, were introduced in the length and at mid-thickness.

Six replicates were produced and tested for each of the specimen types. Load testing was performed on a Mecmesin MultiTest-dV testing bench equipped with a 2500 N load sensor, with a loading speed of 10.0 mm/min. Flexural modulus and ultimate flexural strength were calculated following ASTM D1037. Two specimen series were produced, one by CITA following the protocol in section 3.1, and one by Mogu following the protocol in section 3.3. The wet specimens cultivated by CITA are parallelepipeds of 520 x 72 mm, with a nominal thickness of 20 mm. The width and thickness were not affected by the desiccation, but the length of the dry specimens varied between reinforcement strategies and shrank on average by 3.5 % in the control group, 2.7 % in the inner hessian group, 1.7 % in the hessian jacketing group, and 0.8 % in the rattan group. Mogu cultivated specimens exhibit a volumetric variability of up to 23.8 %. Calculations are taking the dimensional variability into account as averaged per specimen group. Detail pictures of a comparable specimen of each series are presented in Fig.8 and Fig.9.



Figure 8: Detail in top view of CITA cultivated specimen (left), and Mogu cultivated specimen (right).

3.7 Tension series

Following ASTM D1037 dog-bone specimens of a nominal thickness of 20 mm were cultivated by CITA. We report on tension parallel to surface. Our experimental plan investigates the effect of granulate sizes over the tensile modulus and ultimate tensile strength. To this end, two specimen groups were designed:

- Small granulation of beech wood,



Figure 9: Detail in side view of CITA cultivated specimen (left), and Mogu cultivated specimen (right).

- Medium granulation of beech wood.

Six replicates were produced and tested for each of the specimen types. Load testing was performed on a Mecmesin MultiTest-dV testing bench equipped with a 2500 N load sensor, with a loading speed of 4.0 mm/min. Tensile modulus and ultimate tensile strength were calculated following ASTM D1037.

3.8 Simulation model

Simulation of a kagome weave model has been investigated in particle-based simulation in report D5.1. Here, we extend the scope of this study with structural analysis. The analysis was primarily performed with geometrically nonlinear analysis in SOFiSTiK simulation software (SOFiSTiK AG, Nuremberg, Germany). The analysis will use the 3rd principle in SOFiSTiK, it applies all geometric analysis in the setup. The role model is based on the valences-four mixed valences-five carbon fibres demonstration. Our experimental plan investigates the effect of different sizes and different materials for weave members, a the resulting bending strength in the kagome weaves. The material group as follow:

- Carbon fibres strips: Young's modulus of $1.7e5$ for a 20 mm width and 1.5 mm thick member,
- Bamboo strips: Young's modulus of $5e4$ for a 20 mm width and 4 mm thick member.

Thus, we export the hybrid strategy of kagome weaves, from minimum bending radius and analysed the principal stress line of the mycelium shell. Stress-line analysis provide an idealization of material continuity and encode the optimal topology. The strategy is to place the high bending radius material cross singularity. Subsequently, placing high bending strength at high-density region in stress line (Fig.10). Six models of three weaves only models and three integrated mycelium assembly, and simulated in its dead load and 0.2 kN/m^2 on the top. We evaluated the result of the displacement from the analysis report.

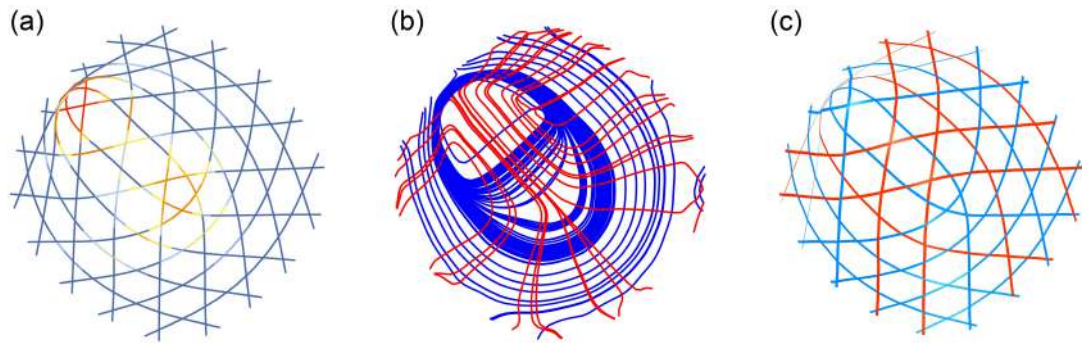


Figure 10: The image of (a) minimum bending radius analysis, (b) principal stress line, and (c) the suggestion of hybrid weaves

3.9 Thermal series

Thermal properties of MBC have been investigated and reported in the literature (see Table 7) with a predominant focus of the measurement of thermal conductivity (λ). A consolidation of results from this literature yields thermal conductivity values in the range of 0.029 – 0.180 W/mK for a wide variety of MBC compositions. Here, we extend the scope of the state-of-the-art by: 1) measuring thermal transmittance (U-value); 2) making preliminary investigations into the thermal properties of material assemblies that suggest construction approaches with MBC. Specifically, we consider a cavity wall typology, with MBC material utilised for both the outer and inner walls. Our experimental plan is designed to firstly assess the thermal properties of MBC panels (approx. 165 x 165 mm after denaturing) of varying thicknesses prepared using *G. lucidum* with two different densities of substrate (Fig.11). This part of the experimental plan comprises the following series:

- Substrate of hemp shives ($\rho = \text{c.}120 \text{ kg/m}^3$); panel thicknesses: 40 mm, 80 mm, 120 mm; six replicates,
- Substrate of birch shavings ($\rho = \text{c.}50 \text{ kg/m}^3$); panel thicknesses: 40 mm, 80 mm; six replicates.

From these results, we calculated the median thermal conductivity of each substrate type and used this to determine the anticipated U-Value of various MBC assemblies. We empirically tested one assembly arrangement (40 mm panel; 20 mm air cavity; 40 mm panel) for both substrate types, with three replicates. We evaluated these results against the predicted performance.

Thermal testing was performed using the gSKIN[®] U-Value Kit (KIT-3610C) which consists of 1 heat flux sensor (XO 67 7C), 1 calibrated datalogger (DLOG-4243) and 2 temperature sensors, to deliver quantitative insulation data (U-Value) according to ISO 9869, ASTM C1046 and ASTM C1155 (Fig.12). These standards cover in-situ thermal transmittance and thermal resistance measurement of building envelopes and components, and are therefore suited to our aim of making preliminary, thermal insulation application focused, evaluations of MBC assemblies for building enclosures.

A thermal chamber was constructed from 100mm rigid insulation panels (Jackodur KF 300 Standard SF100, thermal conductivity (λ_D) 0.035 W/mK) with internal dimensions 500 x 500 x 400 mm (W:D:H). Four sample ports were cut – two ports per face on opposing faces of the thermal chamber – to perform sample tests in parallel, as shown in Fig.13. Each sample required

its own gSKIN[®] U-Value Kit for performing the measurement. The sample ports were cut centred on the mid-point of the vertical dimension of the interior face and distributed with equal spacing across the horizontal dimension of the internal face. The sample port openings were cut at 220 x 220 mm (relative to sample sizes c.175 x 175 mm) to permit packing of insulation between the chamber and the sample.

Thermography was used to verify that there were no thermal leaks at the interfaces between samples and the thermal test chamber. Thermography was conducted using a FLIR One Pro iOS thermal camera (1440 x 1080 pixels, 0.1 ° resolution across a temperature range of –20 to 400 °C). The internal temperature of the thermal chamber was regulated using a heating mat set to 35 °C (Digital Thermostat Thermo 2, Bio Green GmbH, Germany, dimensions 400 x 750 mm, power operation 65 W, max power 3000 W, operating temp. field -50 - 99 °C) in order to guarantee the required minimum temperature delta of 5 °C between internal and external ambient conditions. The heating mat was located in the chamber vertically and equidistant from both sample port faces. The test chamber was located in an unheated basement room with ambient temperature fluctuation between 15-18 °C. Using the proprietary gSKIN[®] software, sample measurements were conducted for a minimum of 72 hours and validated that the U-Value did not deviate more than 5 % from the value 24 h earlier. To make measurements of the 120 mm deep specimens, the thermal chamber was modified with an additional 50 mm of rigid insulation on the faces with port cutouts to mitigate against heat transfer to external ambient conditions from the side of the specimen.

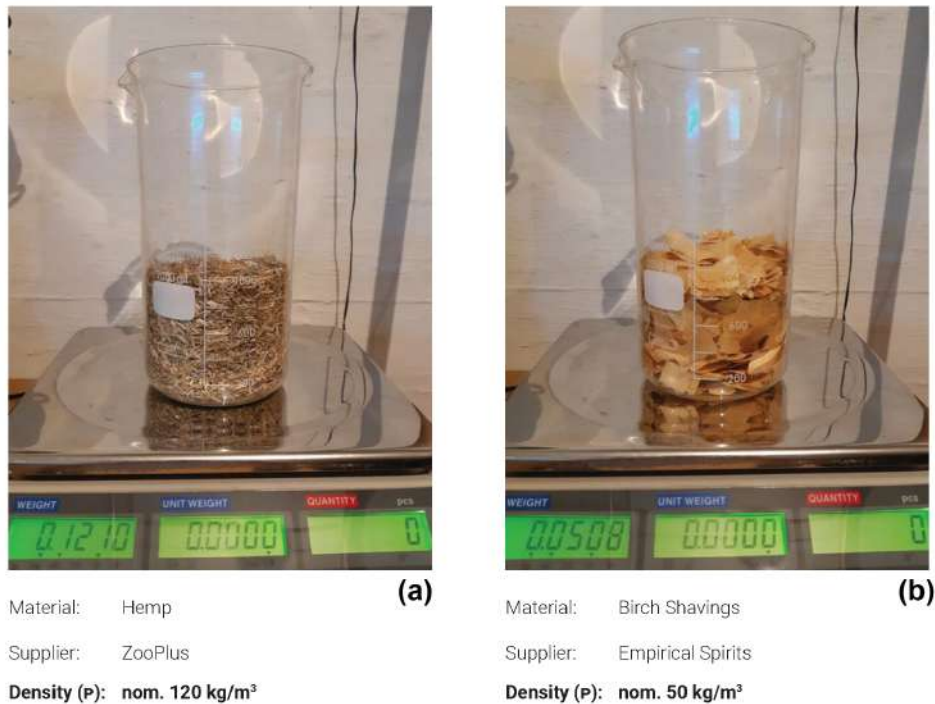


Figure 11: The two substrates used in the production of MBC panels for thermal characterisation – (a) hemp shives, (b) birch shavings



Figure 12: The gSKIN® U-Value Kit (KIT-3610C) used for the thermal transmittance measurements reported here. The kit consists of 1 heat flux sensor (XO 67 7C), 1 calibrated datalogger (DLOG-4243) and 2 temperature sensors. Image source: <https://www.greenteg.com/U-Value/>

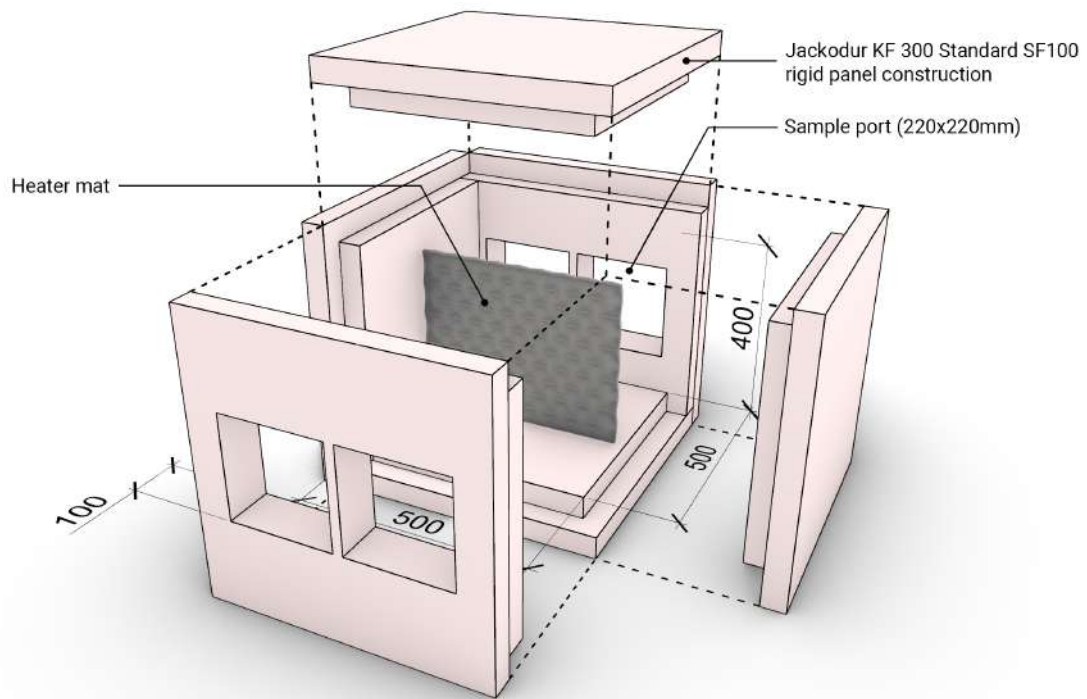


Figure 13: Thermal test chamber constructed to test four samples simultaneously.

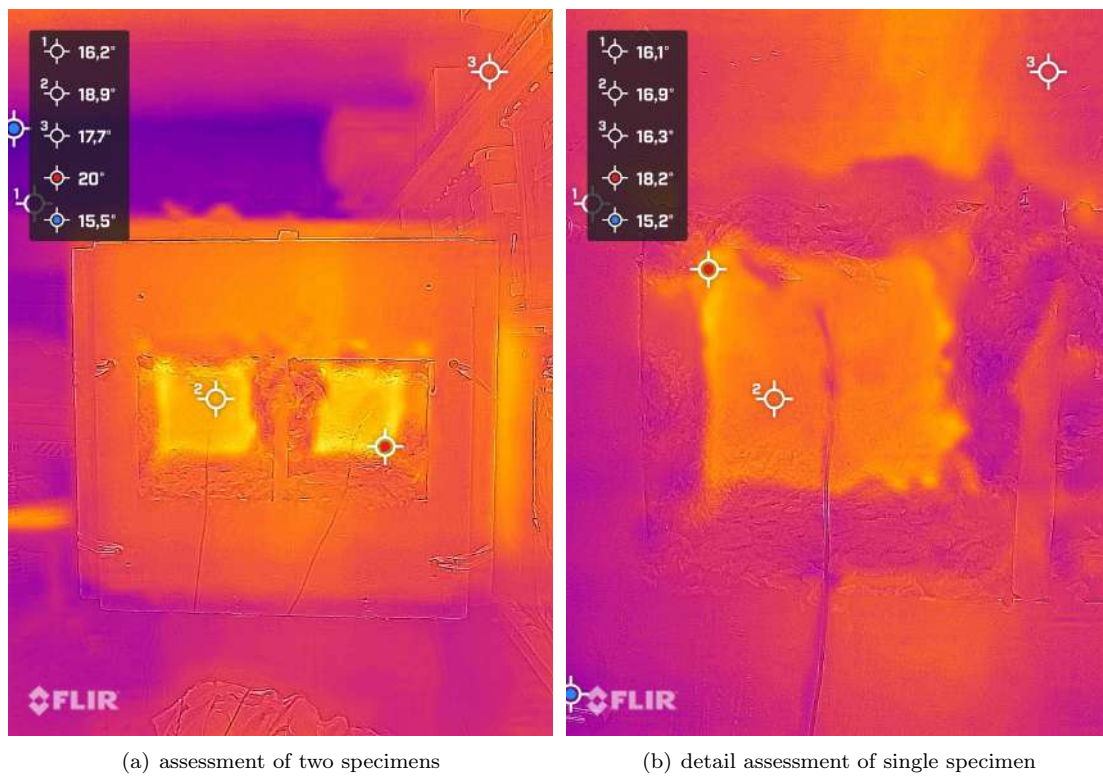


Figure 14: Thermographic verification of the thermal characterisation setup. Two specimens mounted and seen as yellow areas (a), and detail of a single specimen (b), displaying good thermal sealing with only marginal temperature difference between the exposed surfaces and their periphery.

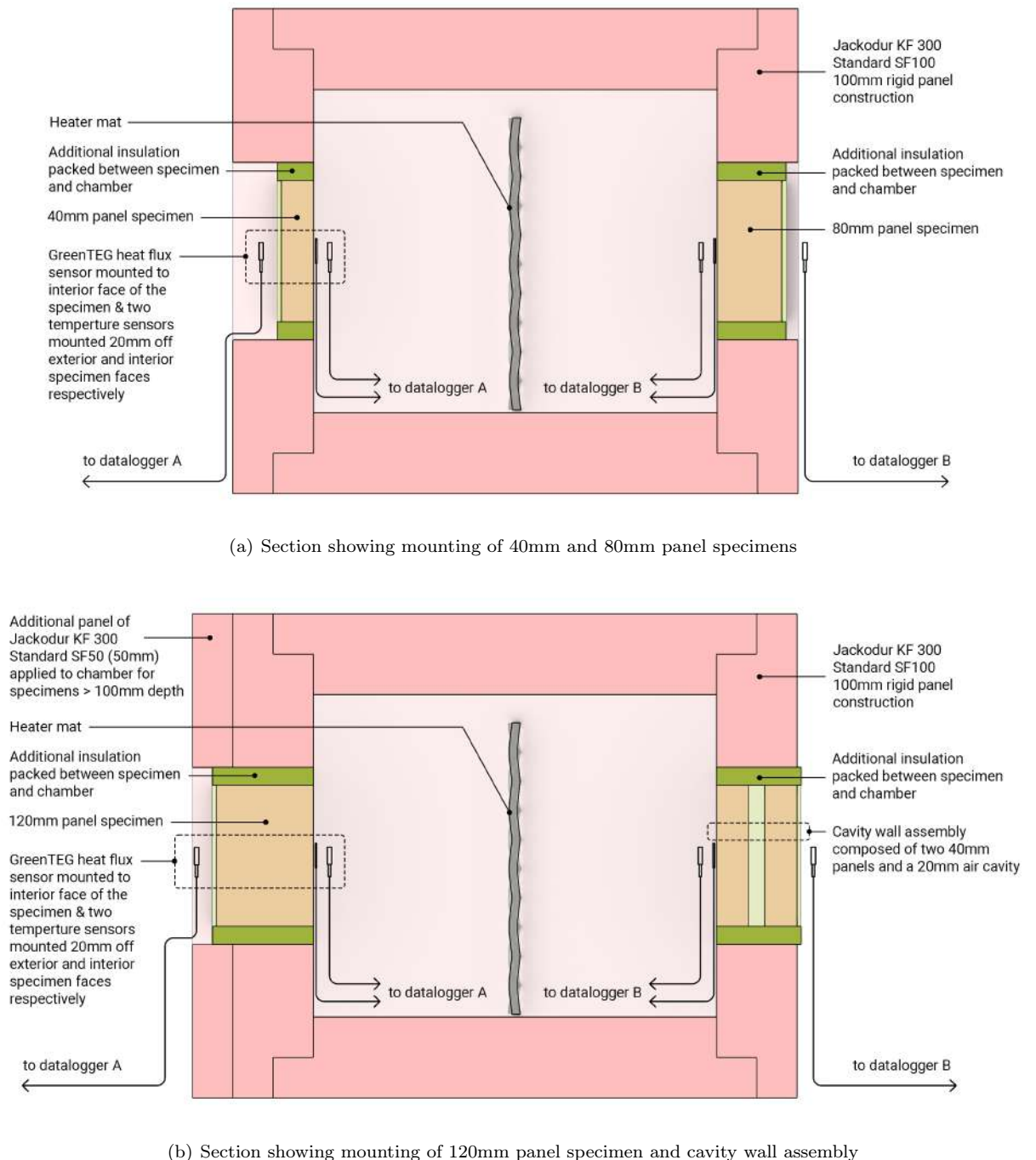


Figure 15: Section through the thermal test chamber setup (a) (b).

4 CHEMICAL CHARACTERISATION

We used FTIR spectrometry to characterise the four raw materials used in the *Compression: effect of particle size & reinforcements* series (Fig.16, section 5.1). The materials were hessian, beech wood, rattan, and common reed. Beech wood and rattan spectra display a chemical profile that is very similar, with the exception of peaks at 1123 cm^{-1} and 1160 cm^{-1} , and the $1300 - 1500\text{ cm}^{-1}$ region. This indicates a slightly higher content of cellulose, hemicellulose and lignin in our tested beech wood specimen (C–O stretching, C–O–C asymmetrical stretching, C–H deformation, COOH groups symmetrical stretching, symmetric C–H bending, CH₂ deformation stretching, CH₃ asymmetrical angular vibration, vibrational mode of amide C–O stretching) [14, 76, 116]. Common reed displays a minimal amount of lignin and hemicellulose compared to our other samples, while the peak at 890 cm^{-1} is associated with C–O–C stretching at the β -(1→4)-glycosidic linkages of amorphous cellulose [27]. The hessian displays distinctive peaks at 707 cm^{-1} , 890 cm^{-1} , 1060 cm^{-1} , 1316 cm^{-1} , and 1430 cm^{-1} in the fingerprint region, and 1640 cm^{-1} , and 2921 cm^{-1} . The $750 - 680\text{ cm}^{-1}$ and $1680 - 1630\text{ cm}^{-1}$ regions (C=O stretching) are associated with primary and secondary amides in hemp (amide V: C–N and N–H vibrations) [120]. Primary amides in hemp are amino acids, fatty acids, and steroids, which contribute to the $3500 - 3000\text{ cm}^{-1}$ region. The $1310 - 1230\text{ cm}^{-1}$ region (C–N stretching) is associated to secondary amides, such as cannabinoids, flavonoids, stilbenoids, terpenoids, alkaloids, and lignans [42]. The peak at 2921 cm^{-1} is associated with alkyl C–H groups [33].

Four samples were isolated from *G. lucidum* colonised beech wood specimens after they were used for load testing. Their spectra were averaged and are presented on Fig.17 along with the beech wood spectrum, and a sample scan of *G. lucidum* mycelium. Peaks at 886 cm^{-1} , 1075 cm^{-1} and 1160 cm^{-1} are characteristic of (1→3)- and (1→6)- β -glucans that are present in the fungal cell wall (identified as [2], and [4] on Fig.17). The peaks [1] and [3] at 780 cm^{-1} and 1043 cm^{-1} are also associated with β -glucans [105]. Chitin is identified at peak [5] 1313 cm^{-1} (amide III: C–N stretching), which also affects the 1640 cm^{-1} region [6] alongside the presence of peptides and secondary metabolites (aromatic rings and conjugated alkenes). The peak [7] at 2922 cm^{-1} is representative of chitin and ergosterol (C–H stretching) [48]. The $3600 - 3000\text{ cm}^{-1}$ region (peak [8]) is considered to be influenced by residual water and entrapped CO₂ (O–H and N–H stretching). Finally, peaks [9] to [12] represent decreases at 1231 cm^{-1} , 1425 cm^{-1} , 1506 cm^{-1} , and 1733 cm^{-1} . They are associated with lignin and xylan breakdown (syringyl ring breathing and C–O stretching, C=C stretching vibration in aromatic ring), and cellulose (peak [11]) and hemicellulose (peak [11] and [12]) breakdown is observed (CH₂ scissor vibration, C=O stretching) [21]. To evaluate the lignocellulosic changes undertaken during *G. lucidum* activity quantitatively, the band ratio indices at 1231 cm^{-1} , 1425 cm^{-1} , and 1506 cm^{-1} were calculated from the 2921 cm^{-1} band [32] for beech wood and *G. lucidum* colonised beech wood as:

$$\frac{I_n}{I_{2921}}, \quad (2)$$

Where I_n is the specific band intensity and I_{2921} the band intensity at 2921 cm^{-1} . The band ratio at 1231 cm^{-1} went from 1.58 in beech wood to 0.61 in *G. lucidum* colonised beech wood; the band ratio at 1425 cm^{-1} went from 0.96 in beech wood to 0.63 in *G. lucidum* colonised beech wood; the band ratio at 1506 cm^{-1} went from 0.71 in beech wood to 0.24 in *G. lucidum* colonised beech wood; the band ratio at 1733 cm^{-1} went from 1.19 in beech wood to 0.41 in *G. lucidum* colonised beech wood. We can therefore observe that *G. lucidum* had a preference in breaking down lignin and xylan at 1231 cm^{-1} compared to cellulose and hemicellulose at 1425 cm^{-1} (2.94:1), which is confirmed by the ratios at 1506 cm^{-1} for lignin (1.42:1), and 1733 cm^{-1} for hemicellulose (2.36:1). The CH₂ scissor vibration corresponding to the peak at 1425 cm^{-1}

reflecting both cellulose and hemicellulose, the present decrease might be primarily related to hemicellulose breakdown. This preference of *G. lucidum* for lignin and hemicellulose is consistent with findings reported in the literature [141].

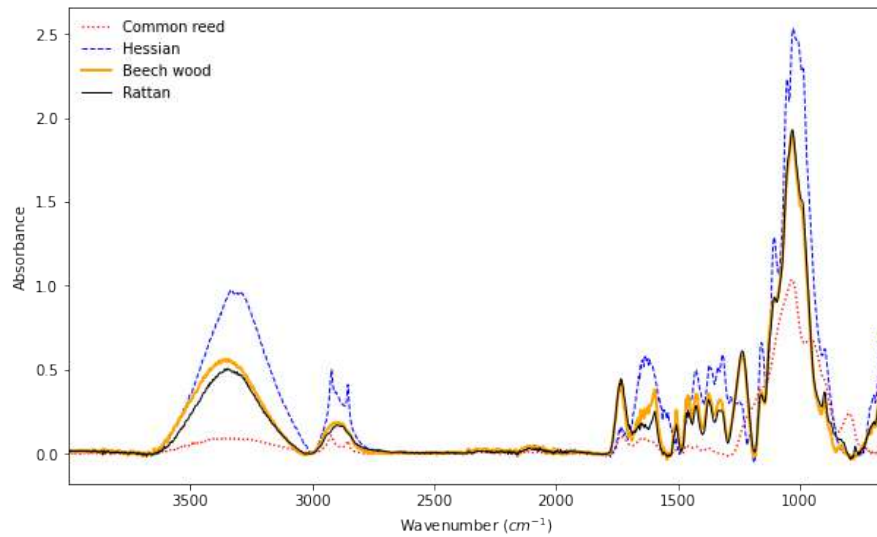


Figure 16: FTIR spectra of hemp-based hessian, beech wood, rattan, and common reed fibres.

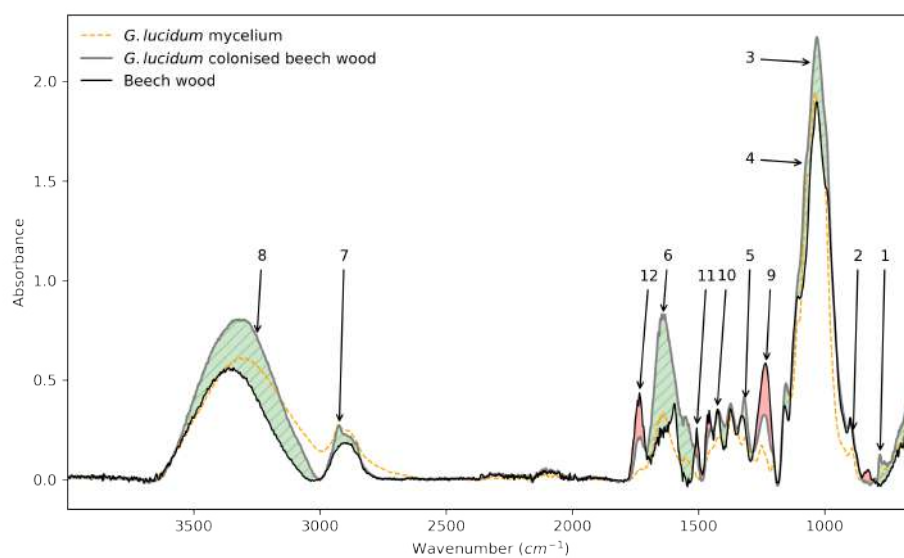


Figure 17: FTIR spectra of *G. lucidum* colonised beech wood, *G. lucidum* mycelium, and beech wood. Green areas represent increased values in mycelium-colonised specimens (peaks 1 to 8), red areas are decreased values in mycelium-colonised specimens (peaks 9 to 12).

5 MECHANICAL CHARACTERISATION

5.1 Compression: effect of particle size & reinforcements

We investigated the effect of particle sizes on the mechanical behaviour in compression of MBC using four levels of granulation: small (BS family), medium (BM family), large particles (BL family), and a 1:1:1 volume ratio mix of the three previous granulations (BSML family). A second parameter was introduced to investigate the anisotropy modification of MBC. Three typologies of fibre composition were implemented in the experimental plan: hessian jacketing coaxial to the load case (H), unidirectional rattan fibres perpendicular to the load case (R), and unidirectional common reed fibres coaxial to the load case (V). Isotropic controls were added for each level of granulation (BS, BM, BL, BSML specimen types in the figures). Fig.18 illustrates the three typologies alongside the control. Experimental parameters per specimen type and resulting mean density, mean Young's modulus and mean ultimate strength are presented in Table 8. Box plots of the results for Young's modulus and ultimate strength are presented in Fig.19. A picture of a specimen from the series is presented in Fig.20, a group of specimens in Fig.21, and specimens after having been tested in Fig.23 and Fig.24.

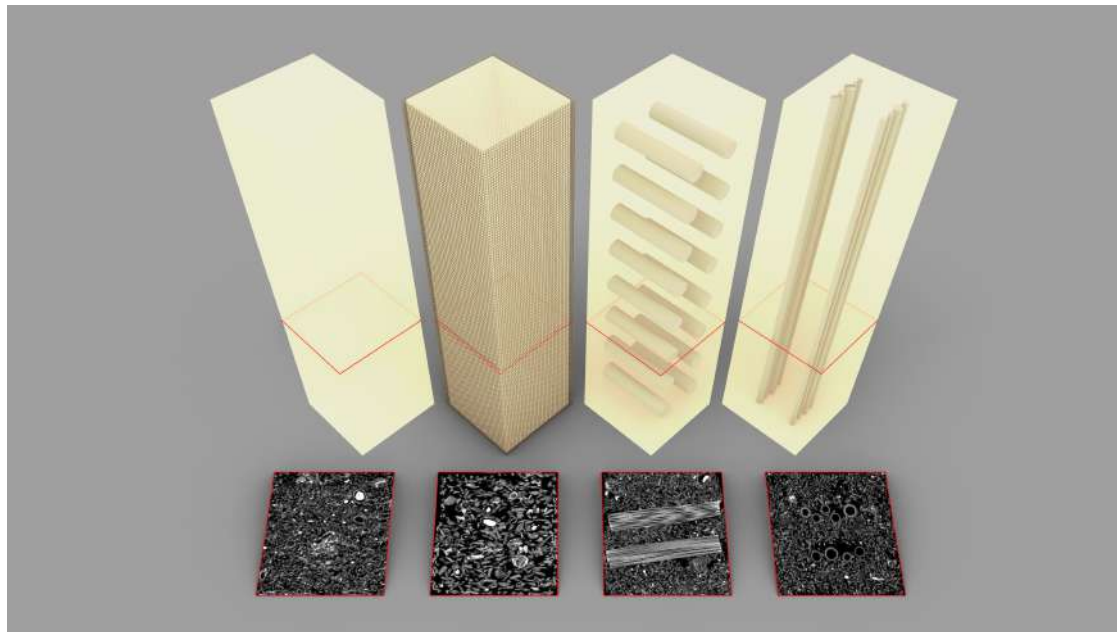


Figure 18: Fibre placement strategies and their sectional CT scan (left to right): control (BS), jacketing coaxial to load (BM_H), fibres perpendicular to load (BS_R), fibres coaxial to load (BS_V).

Jacketing coaxial to load

The introduction of the hessian jacket offers a contrasting illustration of the effect of the mycelial mat usually grown on the external boundary of MBC. We observe that the dispersion of Young's modulus results across all specimen families is reduced compared to their controls with the exception of the BL family. The jacketing also affects the dispersion of results in ultimate

Type	Granulate size	Fibre composition	Mean density (s.d.)	Mean Young's modulus (s.d.)	Mean ultimate strength (s.d.)
BS	0.5 - 1.0 mm	Control	209.67 kg/m ³ (6.47)	1.79 MPa (0.41)	171.86 kPa (36.54)
BS_H	0.5 - 1.0 mm	Hessian jacketing	230.48 kg/m ³ (9.88)	1.58 MPa (0.42)	175.79 kPa (34.38)
BS_R	0.5 - 1.0 mm	Rattan perpendicular	196.59 kg/m ³ (19.01)	0.66 MPa (0.42)	89.06 kPa (58.49)
BS_V	0.5 - 1.0 mm	Common reed coaxial	194.12 kg/m ³ (5.09)	3.88 MPa (2.51)	146.85 kPa (39.26)
BM	0.75 - 3.0 mm	Control	233.87 kg/m ³ (9.04)	3.32 MPa (0.80)	306.38 kPa (57.64)
BM_H	0.75 - 3.0 mm	Hessian jacketing	248.70 kg/m ³ (12.20)	2.99 MPa (0.54)	298.85 kPa (35.47)
BM_R	0.75 - 3.0 mm	Rattan perpendicular	226.77 kg/m ³ (6.19)	4.02 MPa (4.45)	232.30 kPa (62.85)
BM_V	0.75 - 3.0 mm	Common reed coaxial	198.14 kg/m ³ (2.89)	9.21 MPa (6.42)	270.93 kPa (79.76)
BL	4.0 - 12.0 mm	Control	217.60 kg/m ³ (10.58)	2.96 MPa (1.04)	245.60 kPa (30.31)
BL_H	4.0 - 12.0 mm	Hessian jacketing	264.05 kg/m ³ (11.97)	3.01 MPa (0.46)	223.93 kPa (25.88)
BL_R	4.0 - 12.0 mm	Rattan perpendicular	240.98 kg/m ³ (3.91)	2.24 MPa (0.58)	180.88 kPa (64.75)
BL_V	4.0 - 12.0 mm	Common reed coaxial	209.47 kg/m ³ (7.90)	8.50 MPa (4.56)	290.86 kPa (100.83)
BSML	0.5 - 12.0 mm	Control	220.59 kg/m ³ (8.12)	2.17 MPa (0.36)	237.09 kPa (31.73)
BSML_H	0.5 - 12.0 mm	Hessian jacketing	246.85 kg/m ³ (11.29)	2.20 MPa (1.04)	194.48 kPa (48.47)
BSML_R	0.5 - 12.0 mm	Rattan perpendicular	224.09 kg/m ³ (3.55)	1.87 MPa (0.30)	171.44 kPa (26.13)
BSML_V	0.5 - 12.0 mm	Common reed coaxial	203.08 kg/m ³ (6.24)	7.89 MPa (2.41)	338.75 kPa (65.39)

Table 8: Summary of specimen types parameters, resulting dried densities, and compressive properties.

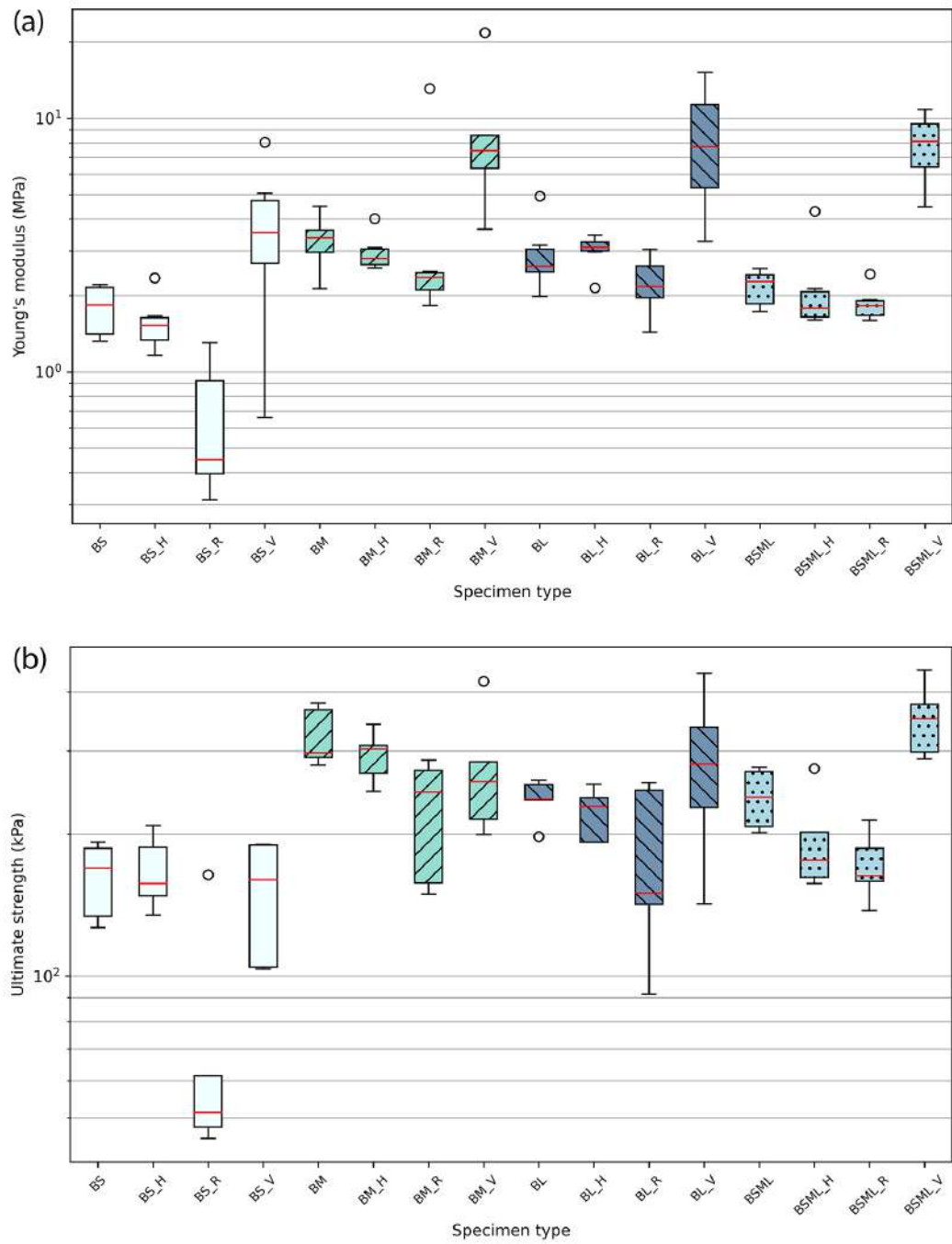


Figure 19: Box plots for compressive Young's modulus results (a) and ultimate strength results (b).



Figure 20: One dried specimen from the *Compression: effect of particle size & reinforcement* series.



Figure 21: Specimens from the *Compression: effect of particle size & reinforcement* series.

strength in the case of the BS, BM, and BSML families, with a reduction of the deviation between the first and third quartiles. The containment of stress applied to the specimens within tight boundaries forces the arrangement of the particles within, restricting the ability for particles to arrange freely. Jacketed specimens have an average reduction of 0.12 MPa to the controls as per Young's modulus (s.d. 0.19), and an average decrease of 16.97 kPa to the controls as per ultimate strength (s.d. 20.05). The jacket has two important advantages: it offers a durable alternative to low-ductility mycelial mats usually grown on the external boundary of MBC, and we hypothesise that it can substantially contribute to an increase in fracture resistance performance in shearing and bending load cases.

Fibres perpendicular to load

Specimens supplemented with rattan fibres display a lower performance across particle sizes considering their median in Young's modulus and ultimate strength. The mean ultimate strength follows the performance of the mean of the controls (Fig.22) with an average reduction of 71.81 kPa (s.d. 8.45). This suggests that, should the production conditions of such MBC improve to reduce the dispersion of results and increase the material behaviour predictability, introducing strategically parsed weakness points in composites could find a use with calibrated materials by tuning their failure mode.

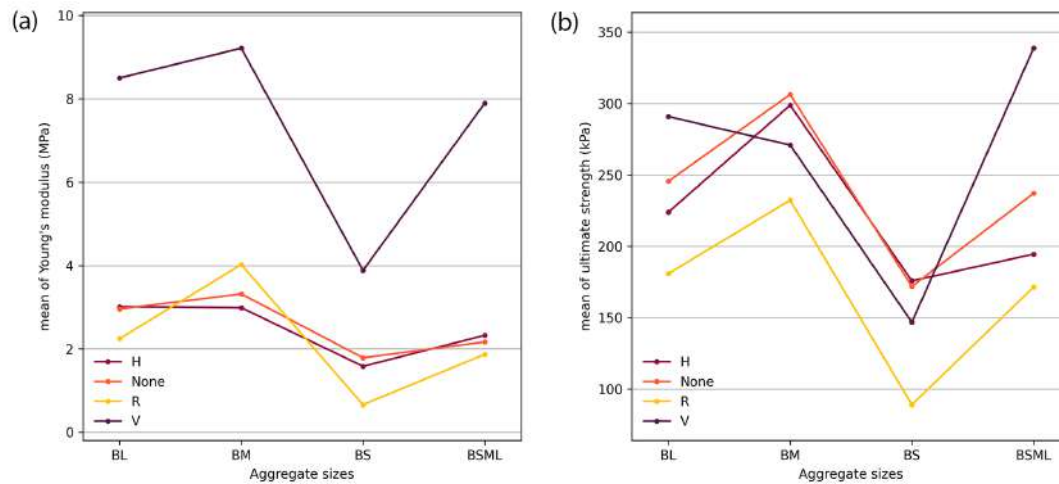


Figure 22: Parameters interaction graph for mean Young's modulus (a) and mean ultimate strength (b).

Fibres coaxial to load

Common reed fibre reinforced specimens resulted in the largest standard deviations in Young's moduli (reported in Table 8), especially in the BM and BL families. This is due to the fibres having partially misaligned to the load case axis during specimen production. Nevertheless, results suggest that MBC can be successfully stiffened with regards to their use case. The effect of this stiffening on the ultimate strength is less obvious as we note that the smaller particles (BS and BM families) tend to perform better without reinforcement coaxial to load. This is a result of the inherent large displacement of the fibres within the specimens under stress due to their

stiffness, thus initiating an early critical failure. The mean Young's moduli (Table 8) display a clear improvement compared to the controls: we observe an average increase of a factor 2.86 (s.d. 0.6) between the mean of the controls and the mean of the fibre coaxial to load specimens. As per mean ultimate strengths, they improved in the BL and BSML families when compared to controls (respectively by a factor 1.18 and 1.43), but decreased in the smaller particles families BS and BM (respectively by a factor 0.86 and 0.88).

Principal substrate particles

The use of smaller particles in MBC increases the surface area to volume ratio of what serves as a nutrient for the fungus, hence facilitating its access to it. Filamentous fungi also need air access to develop a mycelium, thus space between particles, if one desires to have it synthesise a biomass that has a considerable effect over its mechanical properties. The small granulation essentially qualifies as a dust with particles size in the 0.5 – 1.0 mm interval, leaving minimal amounts of air between particles within the constrained boundaries of the specimen mould. The best performing BM family (as per ultimate strength) is composed of 0.75 – 3.0 mm particles, thus embedding particles of a comparable size to the BS dust, while containing particles that are three times as long.

The plastic strain of the composite is contributed to only by particles in such composite, which is clearly exhibited in the range of results in Fig.19. As introduced with the common reed and rattan containing specimens, the dewetting behaviour of the larger particles or reinforcements present in the composite is a principal contributor to damage nucleation. Furthermore, the shape, nature, and distribution of particles in a two-phase composite has been shown to have a substantial influence over the load transfer between members and hence their overall stiffness [115].

Statistical analysis

The result distributions are two-tailed. The mean of Fisher's defined kurtosis for Young's modulus series is -0.3328 (s.d. 0.9353) and -1.0564 for ultimate strength (s.d. 0.5343). Fisher-Pearson's skewness coefficient mean for Young's modulus is 0.5834 (s.d. 0.8540), and 0.1733 for ultimate strength (s.d. 0.5137). The distributions are considered normal [54], which was verified for ultimate strength and Young's modulus results with the Shapiro-Wilk test (respectively $p=0.9224$ and $p=0.0030$, $\alpha=0.001$). Equality of variances was controlled with the Levene test; Young's modulus result variances are not equal ($p=1.3940e-05$, $\alpha=0.05$), neither are ultimate strength ones ($p=0.0459$, $\alpha=0.05$). Welch's ANOVA was conducted for the two parameters: fibre placement for Young's modulus and ultimate strength (respectively $p=0.0001$ and $p=0.0013$), and particle size for Young's modulus and ultimate strength (respectively $p=0.0030$ and $p=4.6462e-09$). The mean values of specimen groups are significantly different ($\alpha=0.005$). Using the pairwise Games-Howell test we identified the most significant reinforcement to be the fibre coaxial to load against fibre perpendicular to load, the control, and hessian jacketing (all $p=0.001$ as per Young's modulus; respectively $p=0.030$, $p=0.004$, and $p=0.004$ as per ultimate strength; $\alpha=0.05$). Continuing this test, we identified the most significant aggregate size to be the 0.5 - 1.0 mm interval (BS family) against the BM and BL families (respectively $p=0.029$ and $p=0.036$ as per Young's modulus; all $p=0.001$ as per ultimate strength; $\alpha=0.05$). The BS family had a significant difference to the BSML family as per aggregate size over ultimate strength ($p=0.001$, $\alpha=0.05$), but not over Young's modulus ($p=0.106$, $\alpha=0.05$).



Figure 23: Rattan embedding specimen used for compressive characterisation, after having been tested.



Figure 24: Hessian jacketed specimen used for compressive characterisation, after having been tested.

Type	Composition	<i>n</i>	Mean density (s.d.)	Mean Young's modulus (s.d.)	Mean ultimate strength (s.d.)
BM_GR	<i>G. resinaceum</i> , BM	5	207.49 kg/m ³ (5.79)	1.75 MPa (0.17)	0.11 MPa (0.02)
PM_GL	<i>G. lucidum</i> , pine wood	2	114.00 kg/m ³ (2.19)	0.34 MPa (0.00)	0.04 MPa (0.00)
BS	<i>G. lucidum</i> , BS	6	209.67 kg/m ³ (6.47)	1.79 MPa (0.41)	0.17 MPa (0.04)
BM	<i>G. lucidum</i> , BM	6	233.87 kg/m ³ (9.04)	3.32 MPa (0.80)	0.31 MPa (0.06)
BL	<i>G. lucidum</i> , BL	6	217.60 kg/m ³ (10.58)	2.96 MPa (1.04)	0.25 MPa (0.03)
BSML	<i>G. lucidum</i> , BSML	6	220.59 kg/m ³ (8.12)	2.17 MPa (0.36)	0.24 MPa (0.03)

Table 9: Summary of specimen types parameters, resulting dried densities, and compressive properties. Values in the lower section of the table are reported from section 5.1.

5.2 Compression: effect of species

We investigated the effect of species on the mechanical behaviour in compression of MBC, with a *G. resinaceum* cultivated on medium granulate size beech wood (Fig.27), and comparing results to the BM group of the experimental series presented in section 5.1. Five replicates could be tested in this group. So as to confirm the lower softwood decaying potential of the *G. lucidum* species, a pine wood specimen group was tested too. Because of a low colonisation ratio as compared to hardwood cultivated composites, only two replicates could be tested for this configuration (Fig.26). Even with a lower number of replicates, it appears that the pine substrate is not a relevant substrate for the *G. lucidum* species. This limited screening is nonetheless consistent with section 2.5 regarding guaiacyl lignins presence in gymnospermous wood and their higher resistance to fungal decay. *G. resinaceum* cultivated specimens resulted in a compressive behaviour similar to small granulation beech wood colonised by *G. lucidum*. The cultivation protocols were identical between these two series. Experimental parameters per specimen type and resulting mean density, mean Young's modulus and mean ultimate strength are presented in Table 9. Box plots of the results for Young's modulus and ultimate strength are presented in Fig.25.

Statistical analysis

Because only two replicates could be tested for PM_GL, this series results are not considered significant and were not statistically analysed. The BM_GR series was tested against the BM series from section 5.1. The result distributions are two-tailed. The mean of Fisher's defined kurtosis for BM_GR Young's modulus is -1.8933 and -1.3775 for ultimate strength. Fisher-Pearson's skewness coefficient mean for BM_GR Young's modulus is 0.0806 , and 0.4800 for ultimate strength. The distributions are considered normal [54], which was verified for ultimate strength and Young's modulus results with the Shapiro-Wilk test (respectively: $p=0.6605$, $p=0.9611$; $\alpha=0.5$). Equality of variances between the BM_GR and BM groups was controlled with the Levene test; Young's modulus and ultimate strength results variances are considered equal (respectively $p=0.0627$, $p=0.0799$; $\alpha=0.05$). Welch's ANOVA was conducted for Young's modulus and ultimate strength (respectively $p=0.0037$ and $p=0.0054$). The mean values of specimen groups are significantly different ($\alpha=0.01$). Using the pairwise Games-Howell test we confirm the significant difference between groups for Young's modulus and ultimate strength (respectively $p=0.004$, $p=0.005$; $\alpha=0.005$).

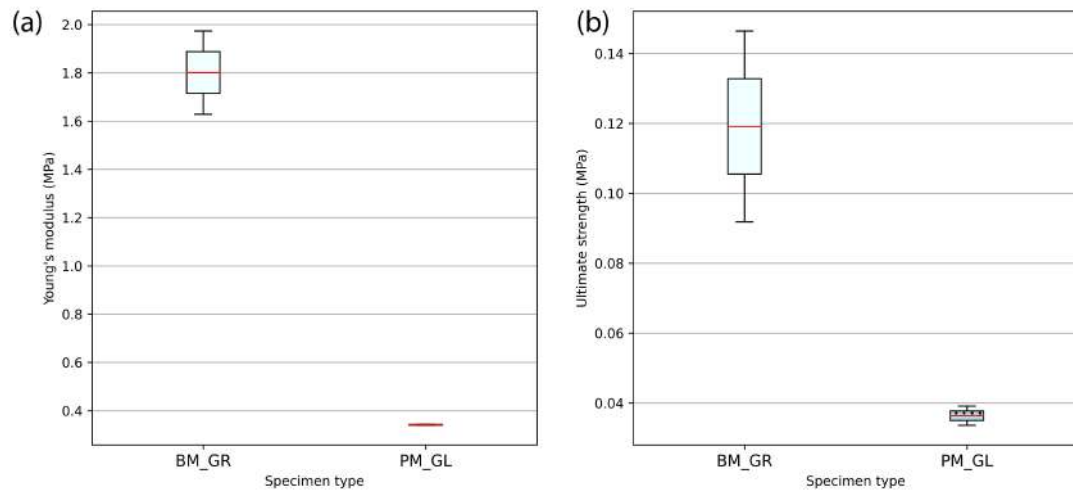


Figure 25: Box plots for Young's modulus results (a) and ultimate compressive strength results (b).

5.3 Flexion: effect of reinforcements

We investigated the effect of diverse reinforcements on the mechanical behaviour in flexion of MBC, using three levels: inner hessian (I), hessian jacketing (H), and rattan fibres in the length (R). These strategies are illustrated in Fig.28, and a picture of specimens is presented in Fig.29. The principal substrate of the specimen is the medium granulation of beech wood (0.75 - 3.0 mm). The BM series was made following protocol CITA, and the HC series was made following protocol Mogu. Isotropic controls were added to the experimental series (BM and HC groups). Experimental parameters per specimen type and resulting mean density, mean Young's modulus and mean modulus of rupture are presented in Table 10. Box plots of the results for Young's modulus and modulus of rupture are presented in Fig.30, and parameters interaction in Fig.31. We can witness that besides rattan, the reinforcements did not have the same effect over the flexural metrics in both series. Considering the significant variability in specimen manufacturing it can hardly be concluded that jacketing or inner hessian can contribute to stirring the composite flexural behaviour in the Mogu series with these experimental data. The mean of Young's modulus for the BM_R series, 1.38 GPa, is the closest in the reported MBC state-of-the-art to industrially viable products for furniture and interior applications, as medium-density fibreboard (MDF) elastic modulus is typically of 4 GPa, with a modulus of rupture of 10 MPa for a density of 750 kg/m³. It can be noted that the mechanical failure of the specimens was related to dewetting of the principal substrate, rattan fibres did not fail nor deform plastically. In Fig.29 we can observe tested specimens crossing at mid-length: they have been softened by aggregates debonding but stayed integral.

Statistical analysis

The result distributions are two-tailed. The mean of Fisher's defined kurtosis for Young's modulus series is -0.8114 (s.d. 0.5732) and -0.5973 for modulus of rupture (s.d. 0.3400). Fisher-Pearson's skewness coefficient mean for Young's modulus is 0.4158 (s.d. 0.6169), and -0.1262 for modulus of rupture (s.d. 0.7371). The distributions are considered normal [54], but did not



Figure 26: One specimen of *G. lucidum* colonised pine wood used for compressive characterisation, after having been tested.



Figure 27: One specimen of *G. resinaceum* colonised beech wood used for compressive characterisation, after having been tested.

Type	Fibre composition	Mean density (s.d.)	Mean Young's modulus (s.d.)	Mean modulus of rupture (s.d.)
BM	Control	216.27 kg/m ³ (17.00)	192.71 MPa (52.40)	0.12 MPa (0.03)
BM_I	Inner hessian	215.12 kg/m ³ (8.01)	197.33 MPa (45.56)	0.11 MPa (0.02)
BM_H	Hessian jacketing	228.77 kg/m ³ (11.60)	375.14 MPa (1.82)	0.18 MPa (0.03)
BM_R	Rattan	245.51 kg/m ³ (9.62)	1.38e3 MPa (630.13)	0.62 MPa (0.14)
HC	Control	123.55 kg/m ³ (2.42)	38.30 MPa (6.72)	0.05 MPa (0.01)
HC_I	Inner hessian	94.74 kg/m ³ (10.96)	20.41 MPa (11.48)	0.04 MPa (0.02)
HC_H	Hessian jacketing	93.49 kg/m ³ (4.98)	11.02 MPa (3.73)	0.03 MPa (0.01)
HC_R	Rattan	103.51 kg/m ³ (6.11)	318.19 MPa (68.50)	0.31 MPa (0.02)

Table 10: Summary of specimen types parameters, resulting dried densities, and flexural properties.

satisfy the Shapiro-Wilk test (modulus: $p=2.1097e-09$, modulus of rupture: $p=2.7600e-07$, $\alpha=0.01$). Equality of variances was controlled with the Levene test; Young's modulus result variances are not equal ($p=0.0005$, $\alpha=0.05$), neither are modulus of rupture ones ($p=9.9545e-05$, $\alpha=0.05$). Welch's ANOVA was conducted for Young's modulus and modulus of rupture regarding reinforcement strategies (respectively $p=0.035$ and $p=0.000009$), and regarding "cultivation media", between CITA and Mogu protocols (respectively $p=0.0071$ and $p=0.043$). The mean values are significantly different ($\alpha=0.05$). Across the two series, using the pairwise Games-Howell test we confirm the significant difference between the rattan group and the other reinforcements for Young's modulus and modulus of rupture ($\alpha=0.05$), besides for inner hessian group against the controls, and hessian jacketing against rattan ($p=0.062$). We confirm the significant difference between cultivation media for Young's modulus and modulus of rupture too ($\alpha=0.02$). In the CITA cultivated series, we confirm the significant difference between all groups for Young's modulus and modulus of rupture ($\alpha=0.05$), but for inner hessian against control.

5.4 Tension: effect of particle size

We investigated the effect of particle size on the mechanical behaviour in tension of MBC, using two granulations: small (BS), medium (BM). The species is a *G. lucidum*. The dog-bone specimen design is presented in Fig.32; the smallest section of the dried dog-bones was of 20 x 36 mm, and 51 mm in length. Experimental parameters per specimen type and resulting mean density, mean Young's modulus and mean ultimate strength are presented in Table 11. Box plots of the results for Young's modulus and ultimate strength are presented in Fig.34.

Statistical analysis

The result distributions are two-tailed. The mean of Fisher's defined kurtosis for Young's modulus series is -1.4313 (s.d. 0.0889) and -1.4840 for ultimate strength (s.d. 0.0013). Fisher-Pearson's skewness coefficient mean for Young's modulus is -0.2065 (s.d. 0.4445), and -0.1546 for ultimate strength (s.d. 0.5246). The distributions are considered normal [54], which was verified for ultimate strength and Young's modulus results with the Shapiro-Wilk test ($\alpha=0.05$). Equality of variances was controlled with the Levene test; Young's modulus and ultimate strength variances are considered equal (respectively $p=0.1011$, $p=0.7427$; $\alpha=0.05$). Welch's ANOVA was conducted for Young's modulus and ultimate strength (respectively $p=0.0030$ and $p=0.0238$). The mean values of specimen groups are significantly different ($\alpha=0.05$). Using the pairwise Games-Howell test we confirm the significant difference between groups ($\alpha=0.05$), but not re-



Figure 28: Fibre placement strategies in the flexion series (left to right): control, inner hessian, hessian jacketing, rattan fibres.



Figure 29: CITA cultivated specimens used for flexural characterisation, after having been tested.

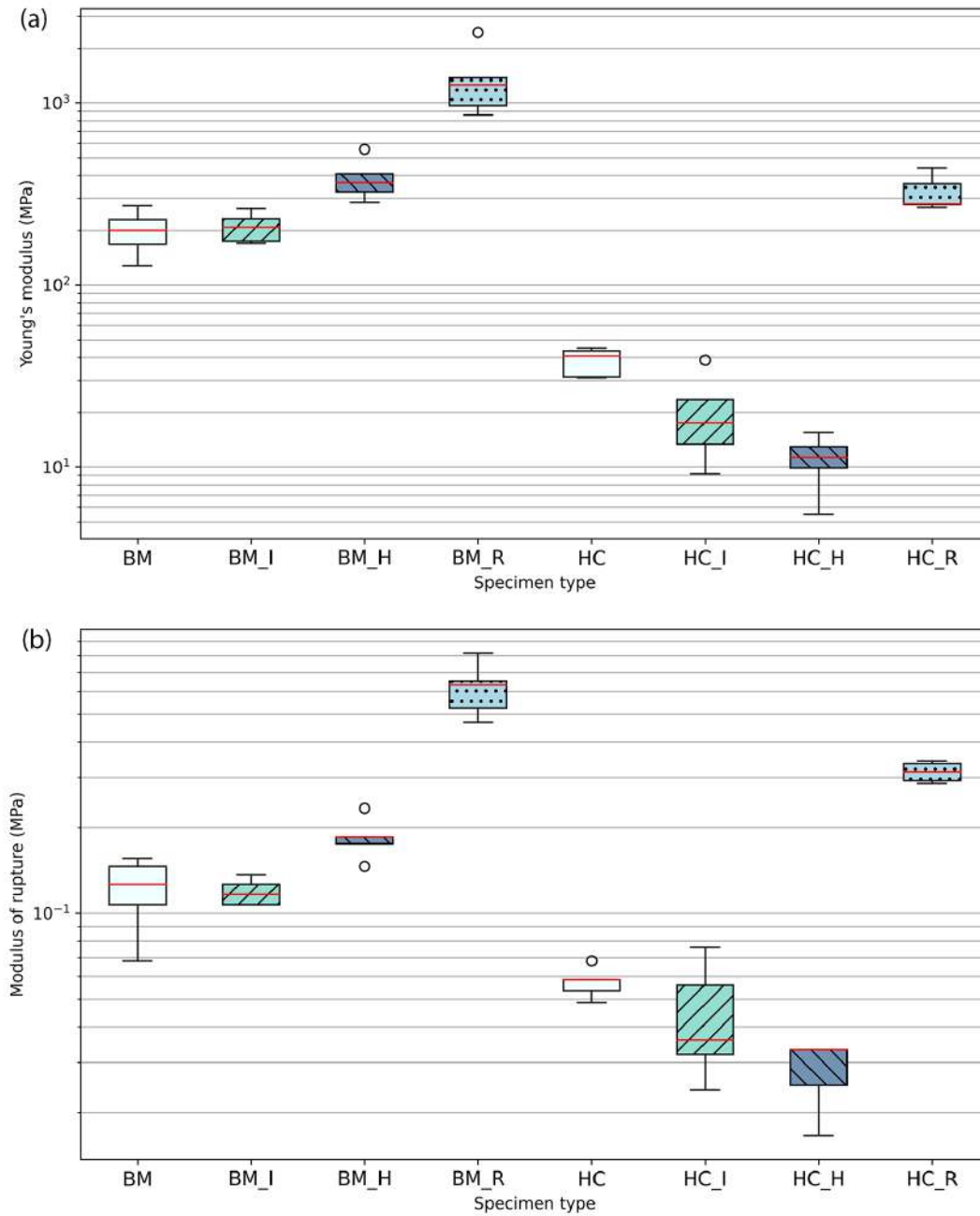


Figure 30: Box plots for Young's modulus results (a) and modulus of rupture results (b).

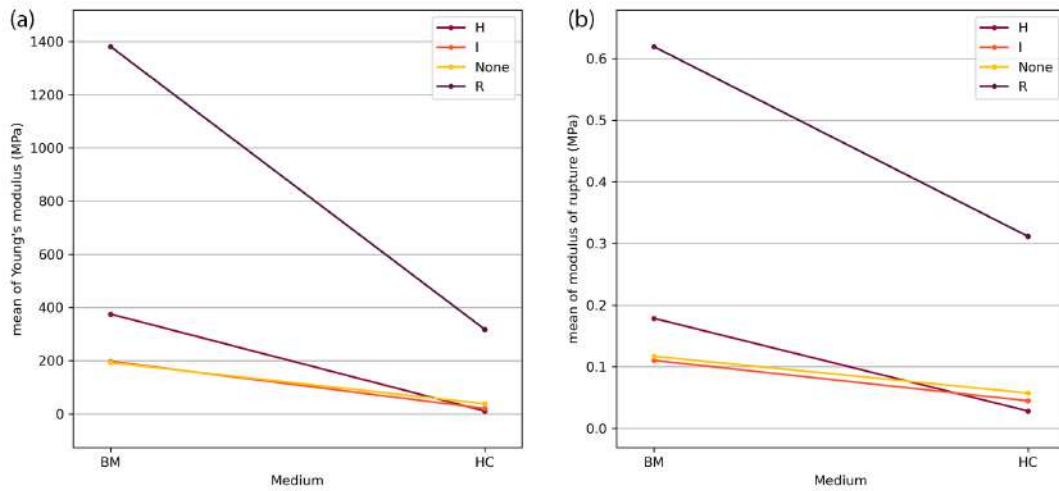


Figure 31: Parameters interaction graph for mean Young's modulus (a) and mean modulus of rupture (b).

Type	Granulate size	Mean density (s.d.)	Mean Young's modulus (s.d.)	Mean ultimate strength (s.d.)
BM	0.75 - 3.0 mm	217.35 kg/m ³ (17.71)	0.84 MPa (0.15)	0.08 MPa (0.01)
BS	0.5 - 1.0 mm	185.15 kg/m ³ (4.82)	0.55 MPa (0.08)	0.07 MPa (0.01)

Table 11: Summary of specimen types parameters, resulting dried densities, and tensile properties.

garding the inner hessian group compared to the control group (both regarding Young's modulus and ultimate strength).

5.5 Kagome structural analysis

We investigated the effect of the structural performance of the kagome Weaves for the mycelium assembly reinforcement. The Strategy is to embed the weaves in the mycelium panels, through the FEM analysis to simulate the structural capacity in the different weave densities of the weaves and the material. The simulation examines two parameters, the types of fibres and the schemes in the kagome. The fibres parameter focused between artificial fibres and natural fibres, carbon fibres, bamboo and rattan. The size is based on the standard industrial dimensions. Experiment the difference of types of fibres is the Young's modulus, the carbon fibres are around $1.7e6$ MPa and bamboo is in the range $4e3 - 6e3$ MPa. The analysis of the kagome was conducted to assess and validate the structural load-bearing capacity of the simulation model. The strips that cross the singularity have high bending and torsion. Due to high bending and torsion, the damage of fibres could either happen during the fabrication or in the different load cases Fig.35. We investigated two approaches for a different condition, one is only to consider the behaviour of weaves without the weight from mycelium assembly. In this setup, the strip elements connect by the cables where the interlacing point, to give some degree of freedom in rotation. The other one integrated the mycelium assembly in one single mesh. Each quad face has been materialised in

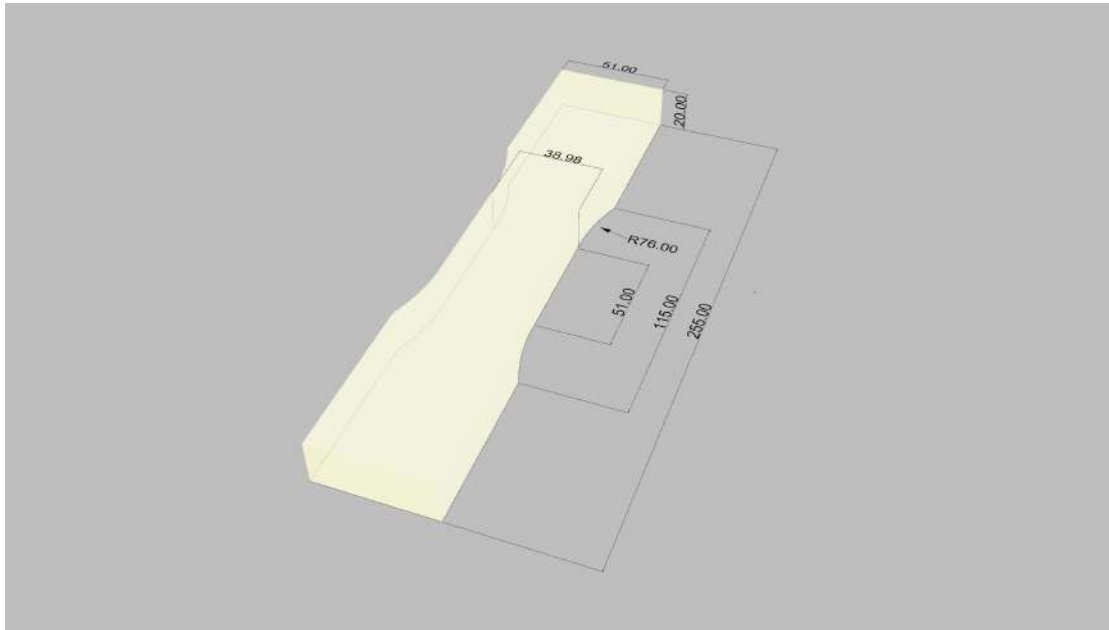


Figure 32: ASTM D1037 informed dog-bone design for the flexion series (wet specimens dimensions).



Figure 33: Two of the specimens tested for tensile characterisation; notice the arrangement of particles and mycelium in the section.

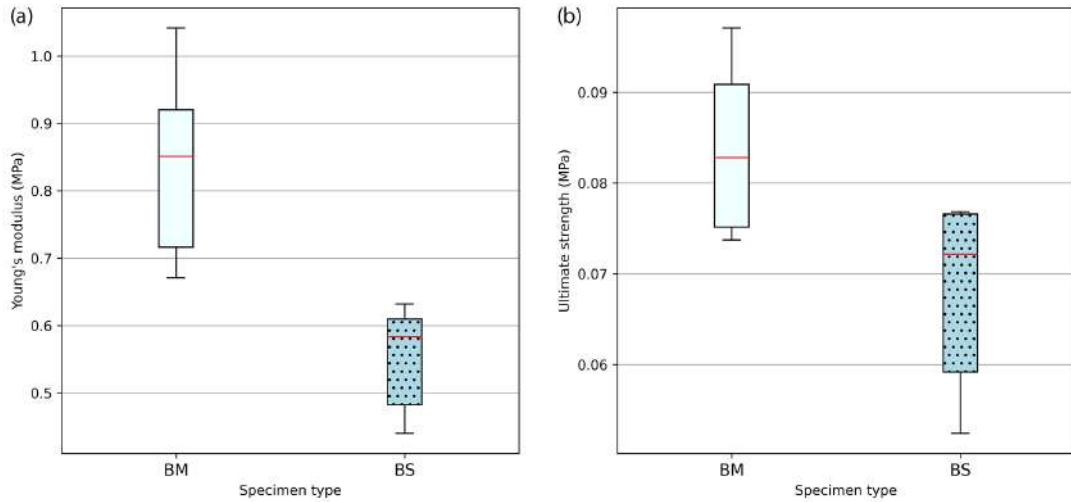


Figure 34: Box plots for Young's modulus results (a) and ultimate tensile strength results (b).

different mechanical properties enable to demonstrate the integration of the kagome weaves and mycelium assembly. The idea for two analyses is to respond to the different stages of construction. The first one is to test the bending and torsion stress, and the second is to understand the behaviour of the final assembly.

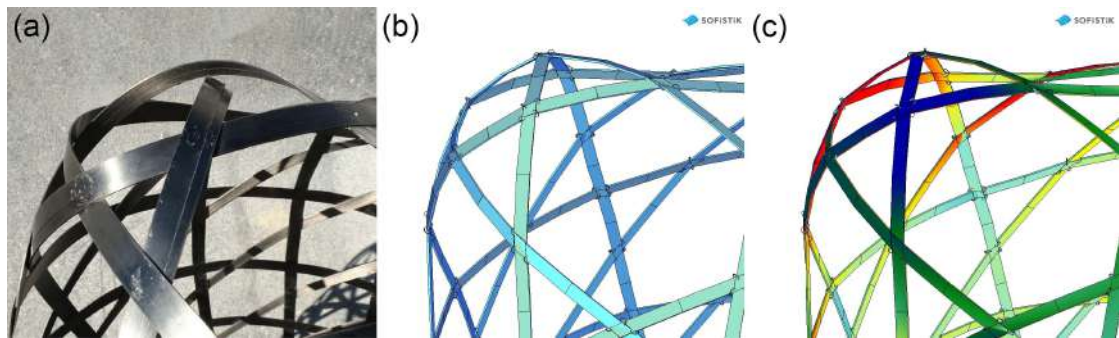


Figure 35: The cable at the intersection constrains the movement and connects two strips.

Structural analysis

Three different materials of the kagome weave and mycelium assemblies could be tested by measuring maximum displacement. At the simulation, the material set is as follows: the bamboo is Young's Modulus $5e3$ MPa, carbon fibres is $1.7e5$ MPa, and mycelium is 17 MPa. In comparison of the kagome weaves with the 0.2 kN/m^2 point load on the top, the maximum displacement: bamboo is 53.1 mm, for the carbon fibres is 9.87 mm and the hybrid is 21.0 mm. The highest von Mises stress surrounds the singularity: bamboo 117.0 MPa, carbon fibres $1.379e3$ MPa, and the hybrid 760.2 MPa (Fig.36). The result confirms the considerable improvement by introducing carbon fibres reinforcement as the kagome hybrid system. It reduced the stress and increased

the load capacity, and lessened the use of carbon fibres.

In the second analysis, the shell model is only considering mycelium use, with a density of 200 kg/m^3 . The model shows a 10.5 mm maximum displacement under its own dead load (Fig.37). Most displacement occurs near the ground support. After applying the kagome substructure, the maximum displacement was reduced to 1.84 mm – 2.05 mm (Fig.38,39,40), with considering the wet density of a mycelium composite (350 kg/m^3). The results contribute to supporting the idea that kagome weaves can be used as a structural framework for a mycelium composite secondary structure (which could be used as an insulation material for instance), and the effectiveness of hybridising the weave with highly performative synthetic composite (carbon fibre) along with natural fibres (such as bamboo) suggests that an efficient distribution could contribute to reinforce structurally demanded areas, while controlling the environmental impact of the architectural system.

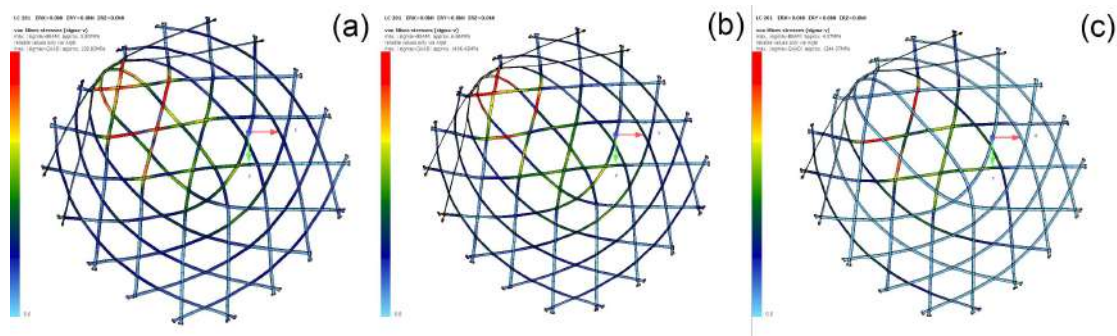


Figure 36: The von Mises stress map of bamboo (a), carbon (b), and hybrid structures (d).

5.6 Conclusion

The result of the experimental series are plotted in Ashby maps along with a curation of state-of-the-art values for compression (modulus: Fig.42, ultimate strength: Fig.43), tension (modulus: Fig.44, ultimate strength: Fig.45), and flexion (modulus: Fig.46, ultimate strength: Fig.47). These maps gather evidences produced with approximately fifteen different fungal species. The series described at section 5.1 is labelled *rigobello_myceliumbased_2021* in the maps, *cita_UnpublishedData_2021* refers to values currently only reported in this document and issued from CITA cultivated specimens. *moguCita_UnpublishedData_2021* refers to values currently only reported in this document and issued from Mogu cultivated specimens.

Both in compression and tension, the effect of particle of the smallest granulation (BS) was significantly lower with regards to ultimate strength and elastic modulus ($\alpha=0.05$). As seen in section 2.3, this may be directly influenced by the presence of air, of cavities between substrate particles for a mycelium to form. Being dust-like, the small granulation (0.5 – 1.0 mm) leave the smallest amount of air within the substrate. Mycelium develops nonetheless, and as the number of particles is higher when their average unit size lowers, we hypothesise that properties of a mycelium interfacing between aggregates plays a more significant role in the artefact mechanical role. Furthermore, the finer grinding of wood aggregates contributes to softening the individual particles; this is due to particles resulting from the shredding process being more elongated on average, and being structurally damaged. A picture of BS, BM, and BL particles is presented in Fig.41. While being consistent industrial products, the granulates display considerable geometrical variations between scales: from small elongated fibres, to flaky planar medium-sized

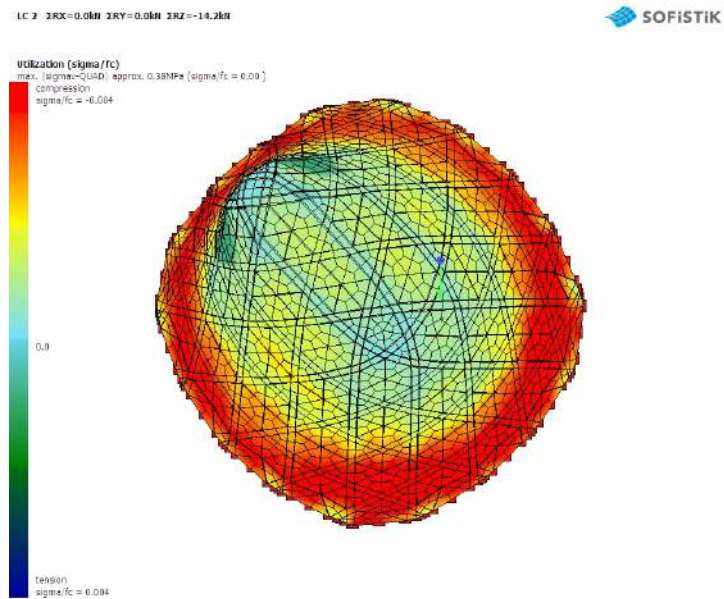


Figure 37: Mycelium shell without kagome reinforcement; compression stress concentrates near ground conditions.

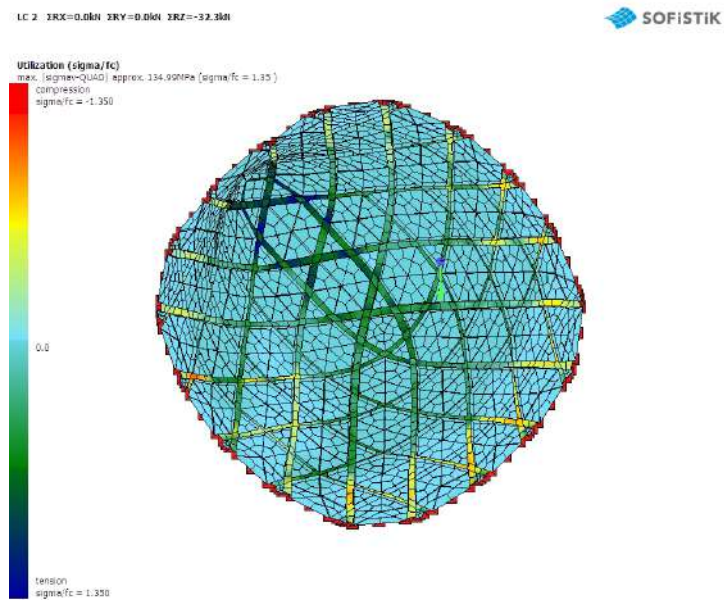


Figure 38: Top view of the bamboo weaves and mycelium simulation in the maximum vertical loading.

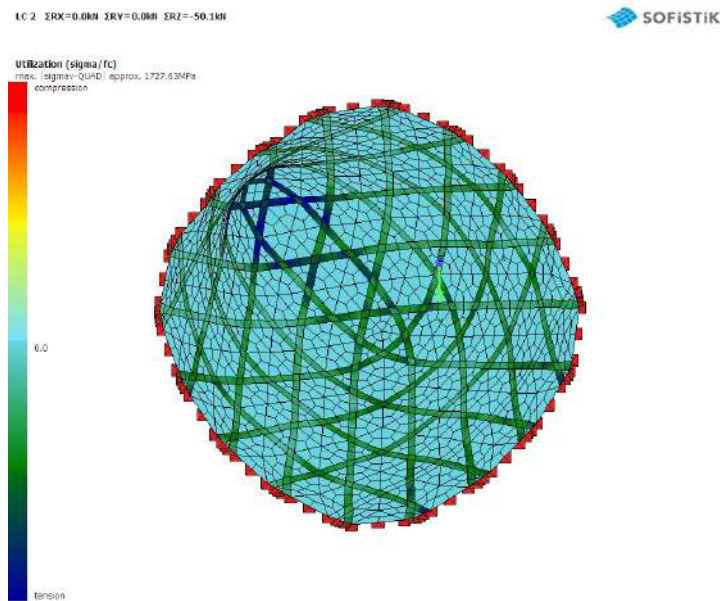


Figure 39: Top view of the carbon weaves and mycelium simulation in the maximum vertical loading.

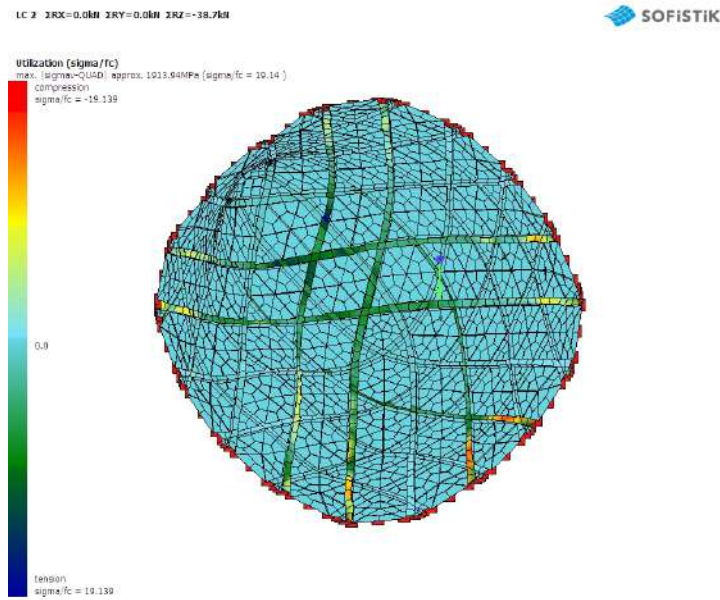


Figure 40: Top view of the hybrid weaves and mycelium simulation in the maximum vertical loading.

particles, to bulky and angular large granulates).

Comparing them to the compression results of BM specimens, we observed that changing fungal species from *G. lucidum* to *G. resinaceum* for an identical experimental protocol resulted in a significantly lower strength and elasticity. To tie this result to section 2.4, a comparative study of isolated mycelia from these two species could be predicted to display a greater elasticity of *G. resinaceum* mycelium, thus expanding the manufacturable material properties. As expected from the enzymatic activity review, and although *G. lucidum* is one of the most versatile selective delignifier, specimens cultivated on pine wood sawdust showed a limited colonisation density and lower performances. Cultivation protocols on gymnospermous woods appear to necessitate changes to ones applied on hardwoods, but the statistical significance of this experiment being non-satisfactory it should be conducted again to be conclusive.

Reinforcement wise, we studied three typologies in compression: rattan fibres perpendicular to load, wheat straw fibres coaxial to load, and hessian jacketing coaxial to load. The addition of fibre coaxial to load had a significant effect over Young's elastic modulus and ultimate strength ($\alpha=0.05$). Because of the wide range in particle sizes used and fibre composition typologies, the significant difference between specimen groups supports our hypothesis that the two-phase particulate model is suited for future MBC studies ($\alpha=0.005$). From this observation, future studies might involve exploring a wider variety of particle shapes, natures, and distributions as these parameters have been shown to have a significant influence over the elastic and plastic behaviour of composites [115]. We demonstrated that the modifying of specimens could be attained with contrasting examples of coaxial reinforcement and perpendicular fracture initiators, with significant effect ($\alpha=0.005$). However, it should be noted that fibre placements were subjected to variability as fibres could partially misalign with the load axis or its perpendicular during production. This suggests that the standard deviation of the results can be reduced by improving the accuracy in production protocol.

Beyond the performances, the various reinforcement strategies in the flexion specimens resulted in different dimensional stability. Across CITA specimens, no shrinkage was measured in thickness nor width, while controls reduced by 3.5 % in length, 2.7 % for the inner hessian group, 1.7 % for the hessian jacketing group, and 0.8 % regarding the rattan group. This is of great practical interest to ensure process control and product quality, and can be integrated and studied further for a variety of geometries. Then both hessian jacketing and rattan fibres introduction contributed to significantly improve the elasticity and strength of composites ($\alpha=0.05$). In the Ashby map Fig.46 we can observe that reinforcements can increase slightly the density of the composites depending on their nature, but nonetheless efficiently increase the stiffness of the specimens (ultimate flexural strength plotted in Fig.47). While the number of represented data points is limited in Fig.46, we can compare the steep increase effect of reinforcements compared to a flatter improvement by a densification strategy from the state-of-the-art [4]. Compressive Young's modulus as a function of ultimate strength is presented in Fig.48. Compressive results that we report in section 5.1 have us observe a steeper increase in stiffness by means of composition as compared to densification strategies [50] (densification can be observed for this data set in Fig.42). Furthermore, we can observe that strategies of supplementation and/or species curating can lead in a performative departure from the state-of-the-art [142]. It can also be noted that no peer-reviewed reproducibility studies have been published yet.

Structural analysis of a mycelium shell and a hybrid kagome structural gridshell have demonstrated the material efficiency interest. Such configurations have only been explored digitally, and will be prototyped in the months to come in parallel to the development of a constructive methodology for embedding computationally active mycelium composites.



Figure 41: Small (BS), medium (BM), and large (BL) granulations of beech wood.

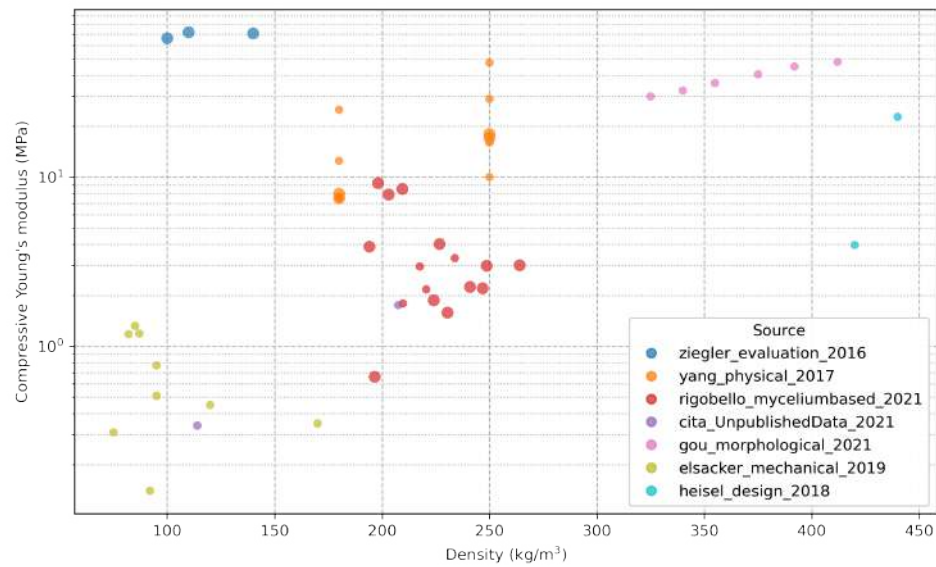


Figure 42: Compressive Young's modulus as a function of density. Data retrieved from [142, 138, 100, 50, 38, 60]. Larger points correspond to composition strategies.

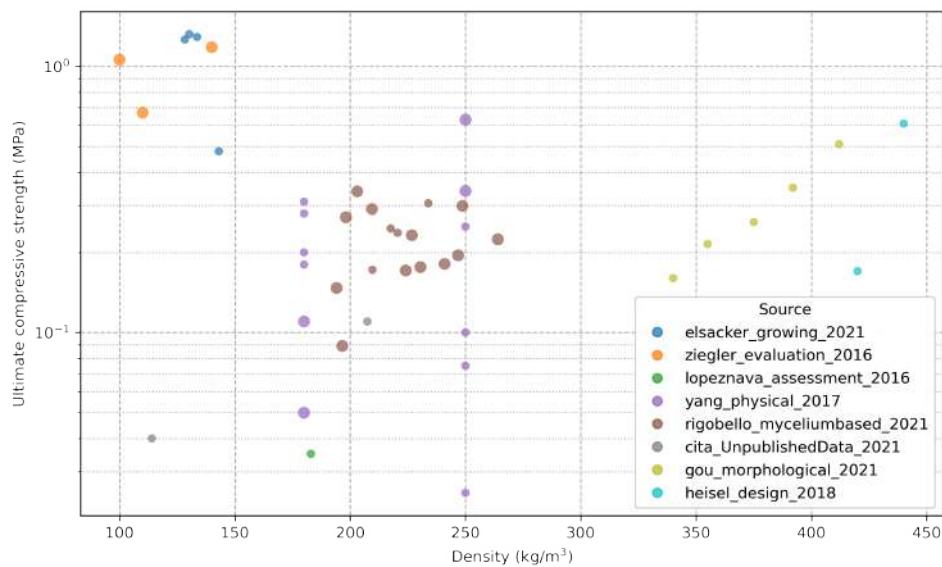


Figure 43: Ultimate compressive strength as a function of density. Data retrieved from [72, 4, 60]. Larger points correspond to composition strategies.

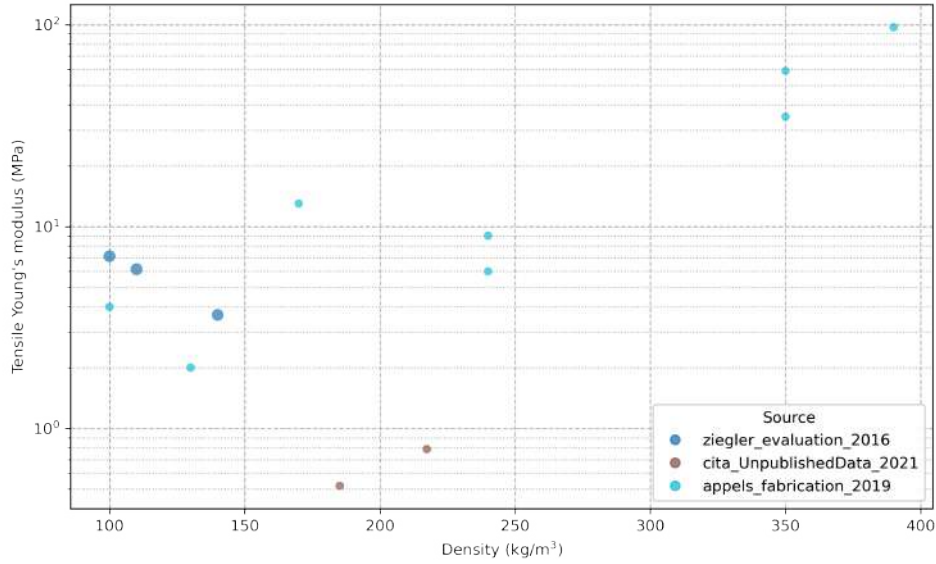


Figure 44: Tensile Young's modulus as a function of density. Data retrieved from [142, 4]. Larger points correspond to composition strategies.

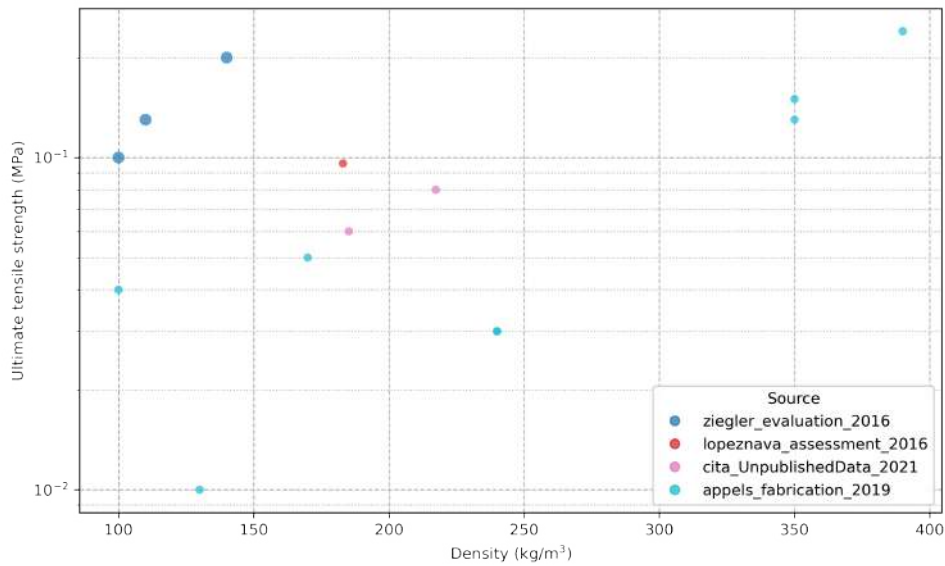


Figure 45: Ultimate tensile strength as a function of density. Data retrieved from [142, 4, 78]. Larger points correspond to composition strategies.

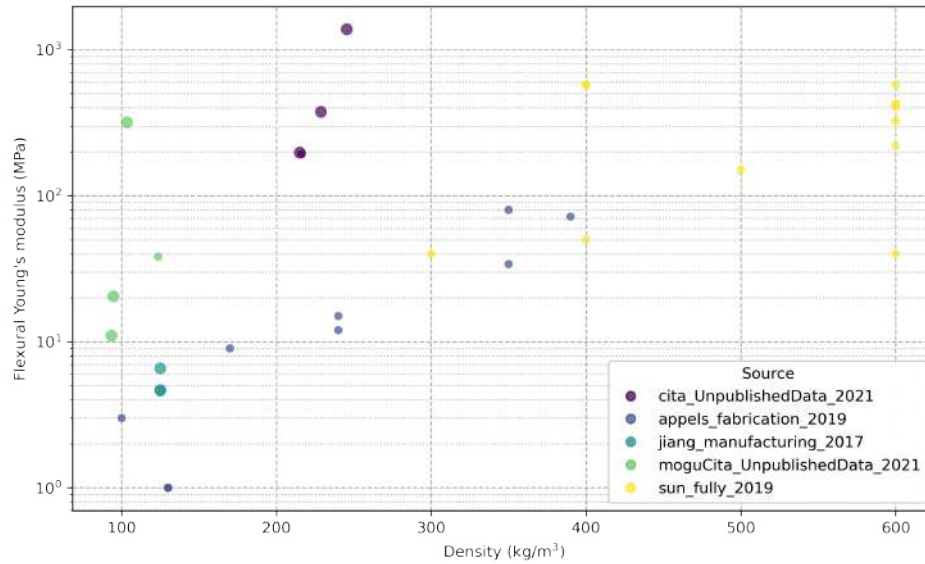


Figure 46: Flexural Young's modulus as a function of density. Data retrieved from [72, 4, 121]. Larger points correspond to composition strategies.

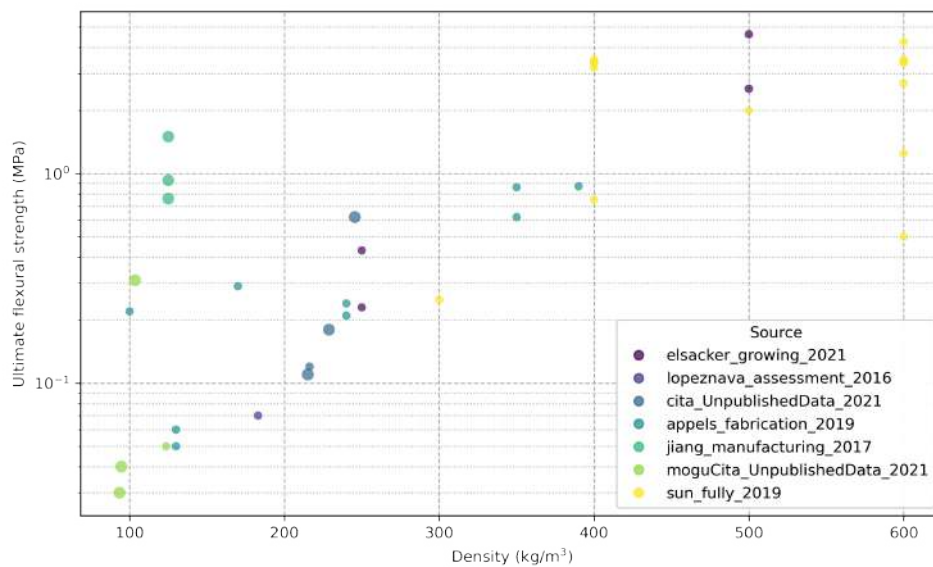


Figure 47: Modulus of rupture as a function of density. Data retrieved from [37, 72, 4, 78, 121]. Larger points correspond to composition strategies.

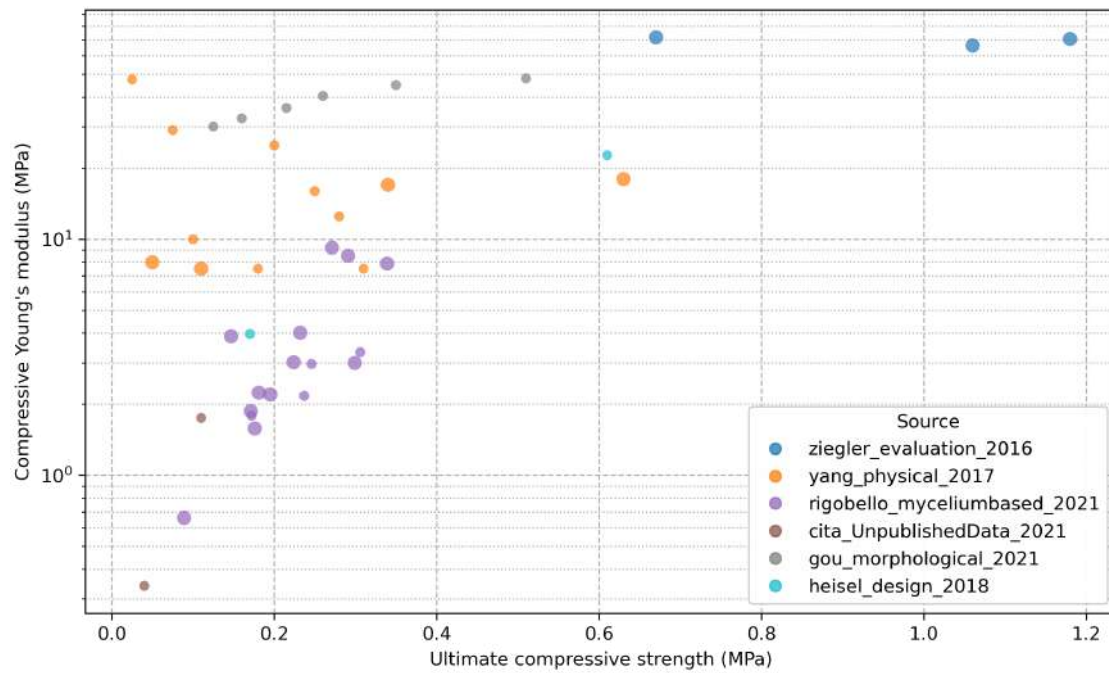


Figure 48: Compressive Young's modulus as a function of ultimate compressive strength. Data retrieved from [138, 142, 100, 50, 60]. Larger points correspond to composition strategies.

6 THERMAL CHARACTERISATION

We investigated the thermal transmittance (U-value) of two series of MBC panels - a hemp shive substrate series and a birch shaving substrate series. Both series were cultivated as described in section 3.2. A composed picture of representative specimens is presented in Fig.49. The hemp shive series consisted of panel thicknesses: 40 mm; 80 mm; 120 mm, with six replicates of each. The birch shaving series consisted of panel thicknesses: 40 mm; 80 mm, with six replicates of each. A custom thermal chamber was constructed to perform U-value measurements on four samples simultaneously. The measurements were taken and analysed according to ISO 9869, as described in section 3.9. The thermal conductivity (λ) was determined from the U-value results for each MBC material series. The thermal conductivity values were then used to calculate predicted U-values for various assemblies of MBC materials. Two assembly configurations were empirically tested and results compared with their predictions to verify their use for design specification.

6.1 Material focused U-value measurement results

For this part of the experimental plan, and in common with studies reported in the literature, we follow a material focused thermal characterisation. However, in contrast to the literature, we measure thermal transmittance (U-value) rather than thermal conductivity or specific heat transfer. Our motivation for this deviation is that the experimental plan extends to consider the composite behaviour of preliminary propositions for assemblies of MBC building envelopes. Therefore, an 'in-situ' measurement methodology is necessary. Conducting all measurements with the same method allows portability of results. Figure 15 illustrates the thermal chamber and measurement setup for different panel thicknesses. Figure 50 presents a comparison of U-value results for hemp shive (H) and birch shaving (BS) substrate series, across 40 mm & 80 mm panel thicknesses and all replicates. Figure 54 presents the calculated thermal conductivity results for H & BS substrate series, across all panel thicknesses and replicates.

6.2 Assembly focused U-value predictions

A cavity wall typology was selected as the basis for making thermal transmittance (U-value) predictions of hypothetical MBC building envelopes. A series of configurations was designed with variations of panel thickness, but keeping consistency of substrate - i.e not mixing H & BS series panels. Figure 51 presents the calculation tables of these preliminary MBC envelope build-ups, together with the predicted U-value results. These calculation tables use the mean thermal conductivity values established from the material tests and presented in Table 12. For comparison, Part L of the UK building regulations (Conservation of fuel and power), stipulates that new build external wall construction must achieve 0.18 W/(m²K) or lower to comply. H_120-20-120 provides the best U-value prediction of 0.20 W/(m²K) for a build-up of 260 mm envelope depth. It is expected that this assembly is predicted to be the best performing as it has the largest insulation depth. The U-value could be improved with an increase in material depth, or an increase in the air-cavity. Very generally, conventional construction of cavity walls will yield a depth of >300 mm. From this prediction of performance, it is plausible to suggest that MBC could potentially result in space saving envelopes.

6.3 Assembly focused U-value measurement results

A cavity wall configuration was implemented for empirical measurement and tested using either H-series or BS-series panels. The configuration was: 40 mm panel (outer face); 20 mm air cavity;

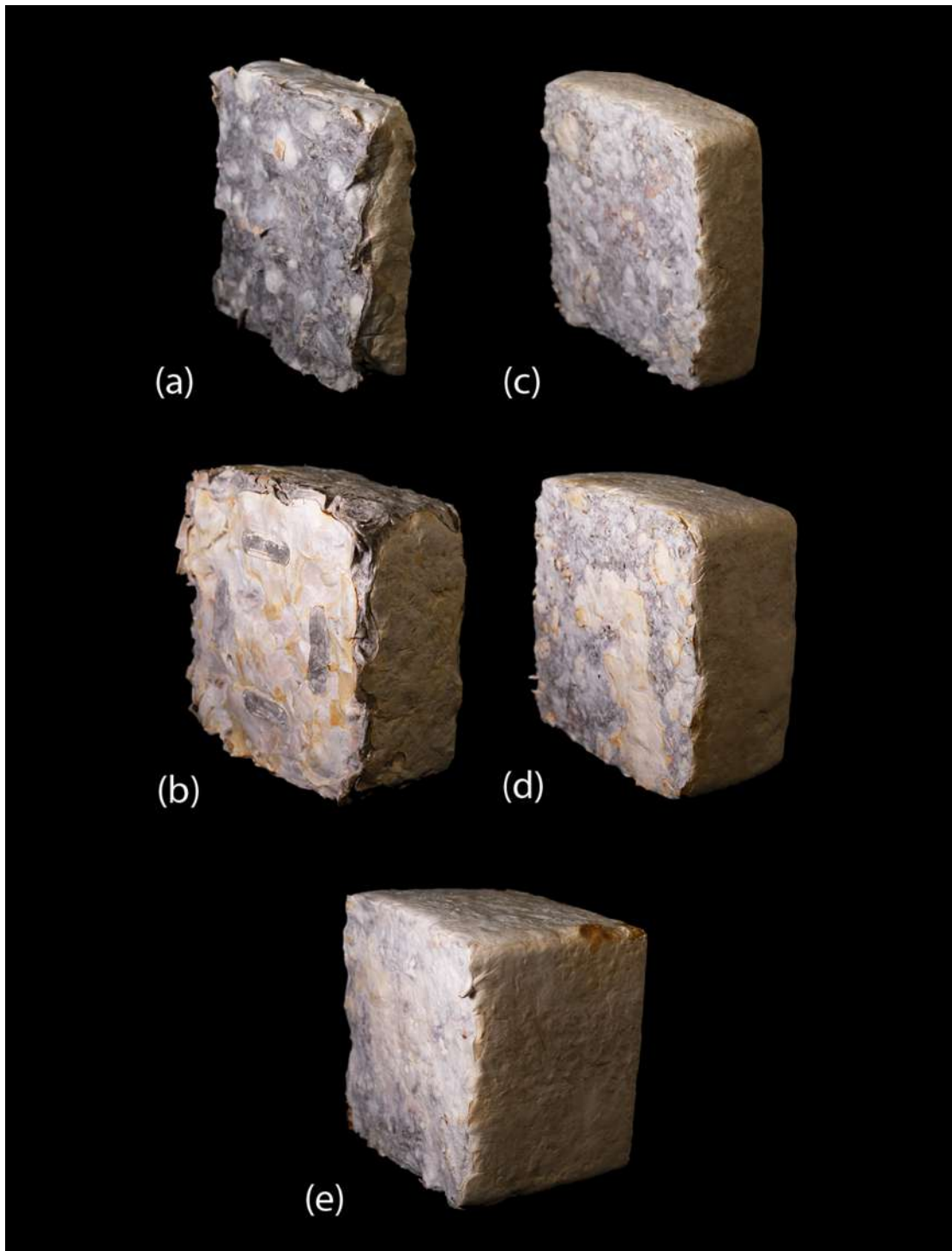


Figure 49: Representative specimens tested for thermal characterisation. 40 mm birch shavings (a), 80 mm birch shavings (b), 40 mm hemp shives (c), 80 mm hemp shives (d), 120 mm hemp shives (e).

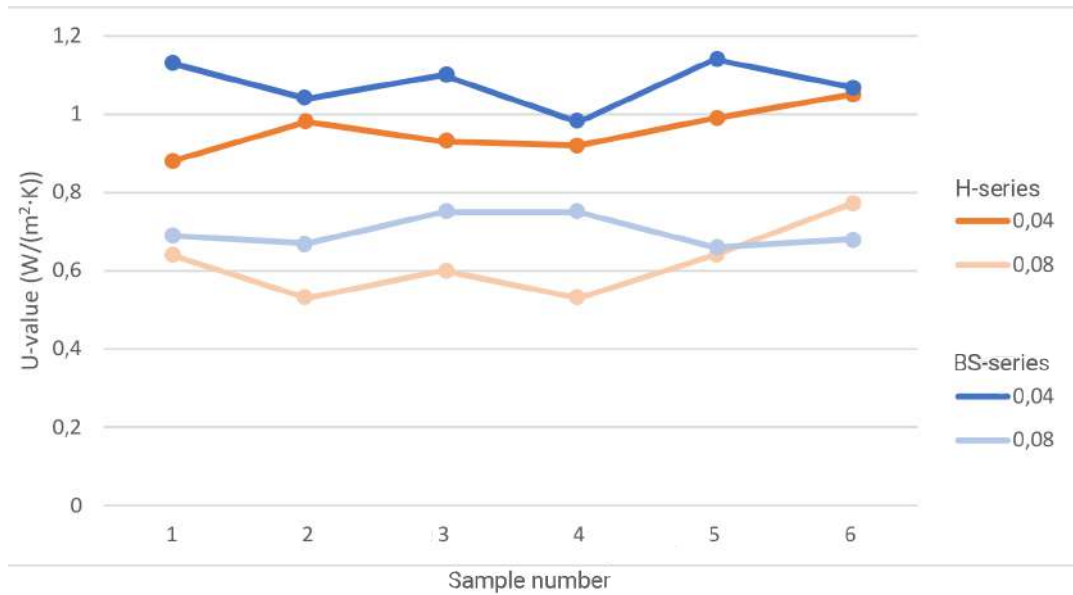


Figure 50: Comparison of thermal transmittance results (U-value) across two material series with two panel thicknesses (0.04 & 0.08m) and six replicates. On average, the hemp series provides better insulation performance.

40 mm (inner face). Three replicates for each material series were measured. The replicates were assembled using panels 1 & 2 for configuration sample 1; panels 3 & 4 for configuration sample 2; panels 5 & 6 for configuration sample 3. Figure 52 presents the measured values and figure 53 presents a comparison of the predicted and measured U-value results. The measured results are substantially higher than their equivalent predictions. Despite the scarce number of data points, the trend is clear and requires further investigation before publication of results.

Statistical analysis

Because only three replicates could be tested for the two buildups (H_40-20-40 and BS_40-20-40), these series results are not considered significant and were not statistically analysed. The result distributions are two-tailed. The mean of Fisher's defined kurtosis for thermal conductivity is -0.9254 (s.d. 0.3667). Fisher-Pearson's skewness coefficient mean is 0.3471 (s.d. 0.5406). The distributions are considered normal [54], which was verified with the Shapiro-Wilk test ($p=0.2846$, $\alpha=0.05$). Equality of variances between specimen groups was controlled with the Levene test; they are considered not equal ($p=0.0348$, $\alpha=0.05$). Welch's ANOVA was conducted; the mean values of specimen groups are significantly different with regards to thickness ($p=0.00007$, $\alpha=0.01$), but not for substrate type ($p=0.8054$, $\alpha=0.01$). Using the pairwise Games-Howell test we confirm that substrate types do not display a significant difference between groups ($p=0.812$, $\alpha=0.05$), while the 40 mm thickness group are significant ($p \leq 0.05$, $\alpha=0.05$), but not 80 mm specimens against 120 mm thick ones ($p=0.332$, $\alpha=0.05$).

Build-up ID H_40-20-40	Thickness	Material	λ	R	Build-up ID BS_40-20-40	Thickness	Material	λ	R
	0,04	H (exterior)	0,04	1,04		0,04	H (exterior)	0,04	0,93
	0,02	Air	0,02	0,87		0,02	Air	0,02	0,87
	0,04	H (interior)	0,04	1,04		0,04	H (interior)	0,04	0,93
				Total R					Total R
				2,96					2,73
				U-Value					U-Value
				0,34					0,37
Build-up ID H_80-20-40	Thickness	Material	λ	R	Build-up ID BS_80-20-40	Thickness	Material	λ	R
	0,08	H (exterior)	0,05	1,62		0,08	H (exterior)	0,06	1,43
	0,02	Air	0,02	0,87		0,02	Air	0,02	0,87
	0,04	H (interior)	0,04	1,04		0,04	H (interior)	0,04	0,93
				Total R					Total R
				3,53					3,23
				U-Value					U-Value
				0,28					0,31
Build-up ID H_120-20-40	Thickness	Material	λ	R		No data for 0,12 BS panels			
	0,12	H (exterior)	0,06	2,05					
	0,02	Air	0,02	0,87					
	0,04	H (interior)	0,04	1,04					
				Total R					
				3,97					
				U-Value					
				0,25					
Build-up ID H_80-20-80	Thickness	Material	λ	R	Build-up ID H_80-20-80	Thickness	Material	λ	R
	0,08	H (exterior)	0,05	1,62		0,08	H (exterior)	0,06	1,43
	0,02	Air	0,02	0,87		0,02	Air	0,02	0,87
	0,08	H (interior)	0,05	1,62		0,08	H (interior)	0,06	1,43
				Total R					Total R
				4,10					3,73
				U-Value					U-Value
				0,24					0,27
Build-up ID H_120-20-80	Thickness	Material	λ	R		No data for 0,12 BS panels			
	0,12	H (exterior)	0,06	2,05					
	0,02	Air	0,02	0,87					
	0,08	H (interior)	0,05	1,62					
				Total R					
				4,54					
				U-Value					
				0,22					
Build-up ID H_120-20-120	Thickness	Material	λ	R		No data for 0,12 BS panels			
	0,12	H (exterior)	0,06	2,05					
	0,02	Air	0,02	0,87					
	0,12	H (interior)	0,06	2,05					
				Total R					
				4,98					
				U-Value					
				0,20					

Figure 51: Predicted U-values of hypothetical cavity wall building envelopes constructed using MBC

Build-up ID	U-values $W/(m^2 \cdot K)$		
	Sample 1	Sample 2	Sample 3
H_40-20-40	0,46	0,53	0,62

Build-up ID	U-values $W/(m^2 \cdot K)$		
	Sample 1	Sample 2	Sample 3
BS_40-20-40	0,59	0,49	0,61

Figure 52: Measured U-values of hypothetical cavity wall building envelope using either H or BS-series MBC 40 mm panels for both interior and exterior wall sections and 20 mm air cavity

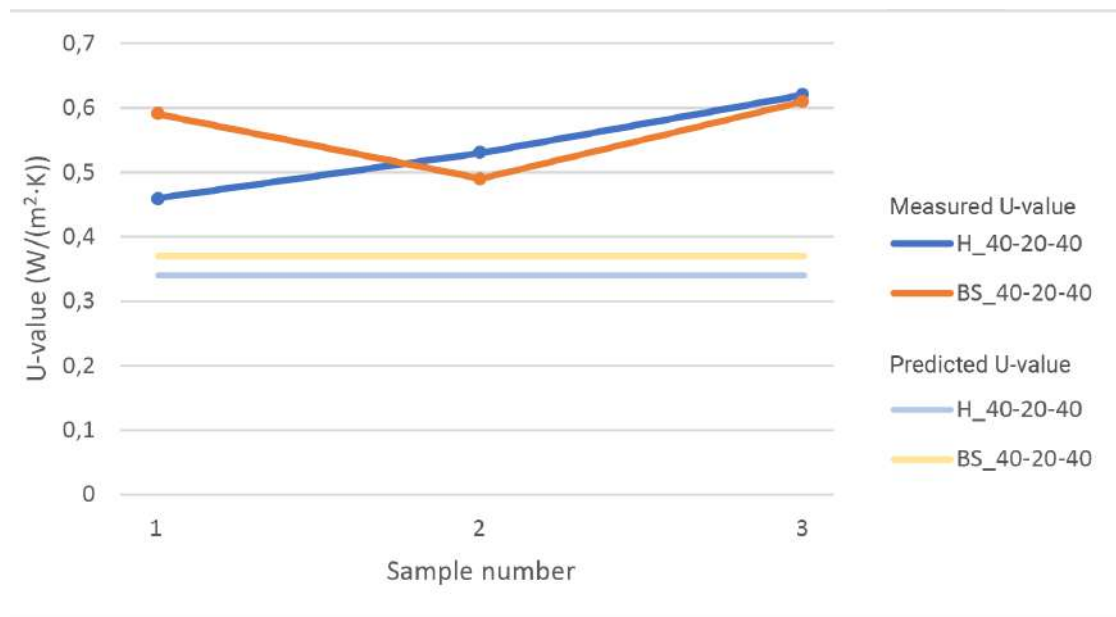


Figure 53: Comparison of predicted and measured U-value results for the tested assembly using two different MBC substrate series.

Type	Substrate	Total thickness	<i>n</i>	Mean density (s.d.)	Mean U-value (s.d.)	Mean λ (s.d.)
H_40	Hemp shives	40 mm	6	107.30 kg/m ³ (4.85)	0.9583 W/m ² K (0.0604)	0.0383 W/mK (0.0024)
H_80		80 mm	6	93.21 kg/m ³ (2.23)	0.6183 W/m ² K (0.0893)	0.0495 W/mK (0.0071)
H_120		120 mm	6	85.47 kg/m ³ (3.83)	0.4867 W/m ² K (0.0665)	0.0584 W/mK (0.0079)
BS_40	Birch shavings	40 mm	6	96.97 kg/m ³ (1.52)	1.0767 W/m ² K (0.0602)	0.0431 W/mK (0.0024)
BS_80		80 mm	6	87.76 kg/m ³ (8.32)	0.7000 W/m ² K (0.0400)	0.0560 W/mK (0.0032)
H_40-20-40	Hemp shives	100 mm	3	-	0.5367 W/m ² K (0.0802)	0.0537 W/mK (0.0080)
BS_40-20-40	Birch shavings	100 mm	3	-	0.5633 W/m ² K (0.0629)	0.0563 W/mK (0.0064)

Table 12: Summary of specimen type parameters, number of replicates and thermal properties for each type.

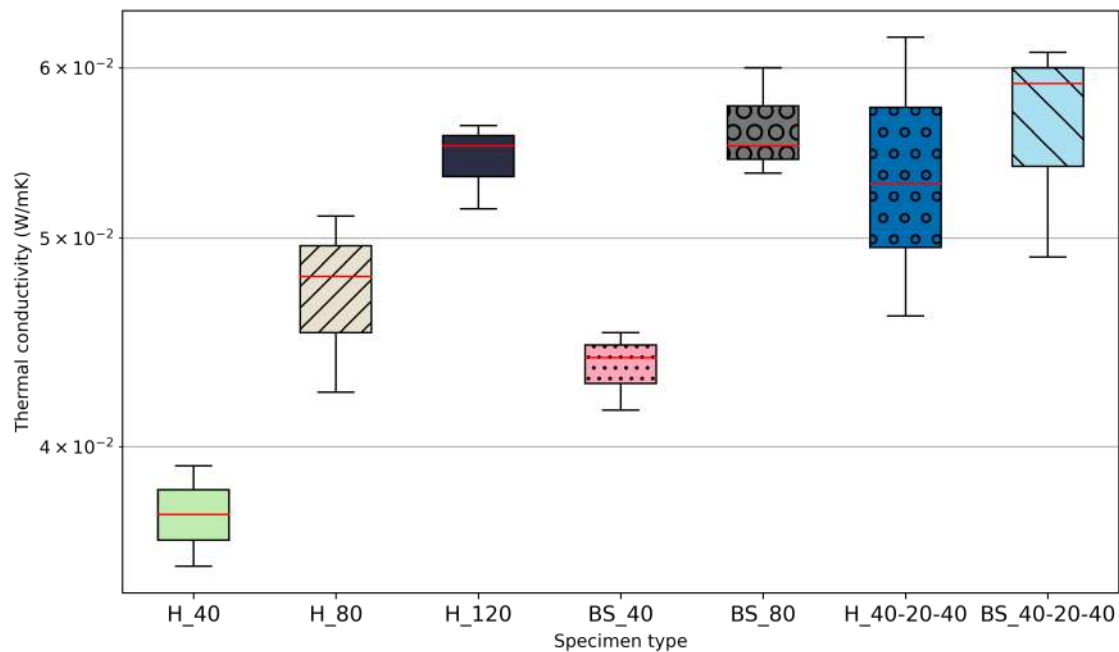


Figure 54: Box plots for thermal conductivity results.

6.4 Conclusion

According to DIN 4108 *Thermal insulation and energy saving in buildings*, materials must have a thermal conductivity below 0.1 W/mK to be defined as insulation materials [107]. Our material focused results show that all panel specimens across both material series qualify as insulation materials, with thermal conductivity values λ in the range of 0.035 - 0.072 W/mK. This is compared to orthodox insulation materials used in construction such as glass wool (0.04 W/mK), mineral wool (0.04 W/mK), XPS foams (0.029 - 0.032 W/mK) and EPS foams (0.030 - 0.040 W/mK). This suggests that, with further improvements in production, MBC could act as a plausible alternative to current mineral and oil based insulation materials. Improvements might be sought through additives to the substrate to further reduce both conductive and radiative heat transfer. It is generally expected that lower density materials should act as better insulators. It is therefore notable that for equivalent panel thicknesses, the H-series out-performs the BS-series (Fig.) despite being composed of a higher density substrate. Causality has not been investigated further, but we hypothesise that geometric features of internal morphology (e.g air pocket volume and geometry, wall thickness of pockets, etc.) could play a role on thermal performance, as has been reported in the literature [93]. The results from the assembly tests are limited, with only three data points per assembly configuration. To increase this dataset we can perform a combinatorial mixing of panels which will yield an additional 12 data points, per configuration. This is being implemented in advance of planned publication. In addition, the inclusion of primary structural elements, connectors and surface finishes should be considered for a more complete assessment of the composite behaviour of MBC based building envelopes.

7 CONCLUSIONS AND PERSPECTIVES

From the enzymes review we come to understand that resulting substrate fitness and mycelia mechanical behaviour can be adjusted by means of supplementation. This can be approached by pH and temperature regulation, and sugars, nitrogen and metal ions supplementation. The properties of mycelia have furthermore been shown to be influenced by such. Use of compounds such as pH buffer CaCO_3 could be systematically investigated in the future. Lignocellulosic substrates selection can also have a significant effect, unsupplemented softwoods being theoretically not chemically fit for optimal white-rot decay. Use of sycamore or ash wood could be investigated in the future as they display higher syringyl monomer contents. The C:N content of sycamore being 401, supplementation may be necessary for optimal fermentation.

The effect of water activity upon MBC colonisation-related performances, such as mechanical response, has not been reported yet in the state-of-the-art. Considering reports in SSF studies of several folds increases in colonisation rates, investigating its effect in MBC may result in improved lead production times and performances. Moreover, qualifying the effect of hydroxyl groups availability in hemicellulose and cellulose depending on previous wood processing could lead to further medium optimisation. We expect this particular aspect to be specifically useful for paste-like substrate design, such as found in mycelium-related 3D printing.

Of course, stirring the focus of future researches towards such specific aspects may result in a higher barrier to entry of a technique that has a strong vernacular interest. The use of FTIR spectrometry by experts within a network of amateur practitioners could be useful to adapt cultivation strategies in a constructive manner: the qualification of an available substrate may lead to specifying necessary processing for it, supplementation, species selection, and/or relevant composition strategies. While industrial and academic researches may result in defining a panel of protocols for a selection of performative materials, constructive practices informed by locally available materials could coexist.

To the best of our knowledge, we report for the first time on particle geometries effect on mechanical properties of MBC. Because the cohesion between them is critical to composite stiffness, we expect the state-of-the-art to benefit from systematically investigating the shape, size, and orientation of these. This may lead in the future to the emergence of heterogeneous materials with gradients of substrate qualities. Such development may lead to material optimisation in design, and creative explorations.

Similarly, the investigation of natural fibres composition in the substrate has proved to lead to significant increases in stiffness and ultimate strength. Composites manufacturing accuracy improvements are expected to contribute to reducing the standard deviation of results. Furthermore, reinforcements may be strategised in developing efficient or multi-functional composites, for instance, in designing the principal substrate to be thermally performative and introducing reinforcements to perform structurally. Such strategy could be used to mitigate manufacturing risks: if a composite is subjected to production inconsistencies, its composition may increase its stiffness and user safety. Moreover, the mean of Young's modulus for the BM_R series, 1.38 GPa, is still close to three times lower than that of MDF (4 GPa), and approximately 16 times lower in modulus of rupture (0.62 MPa, MDF: 10 MPa). Considering the higher density of MDF, 750 kg/m³, as compared to the BM_R series (245.51 kg/m³), a future investigation could be a hybrid strategy of composition and densification so to improve aggregates interlocking and the resulting strength. The arming strategy adopted with the introduction of inner hessian did not prove significant, as it was positioned at mid-height of the specimens it was theoretically not subjected to tension and could only partially contribute to the bending stiffness. A next iteration of the experiment may look into different soft arming positioning strategies by reflecting on the most mechanically demanded areas of the specimens. This elastic modulus, to the best of our

knowledge, is the highest reported for MBC, hence the closest to industrially viable products for furniture and interior applications. We report on various specimen dimensional stability after drying linked to diverse reinforcements, this aspect can be investigated in the future both as a means to explore its design consequences, and for production control.

We analysed the plotting of various data sets from the state-of-the-art along with data from this report against the various tuning strategies (supplementation, densification, and composition). Popular densification techniques, happening at substrate preparation or by cold or hot-pressing, display a larger increase in ultimate strength than stiffness compared to composition techniques. We have experimented and reported the effect of various fibre addition: the use of rattan fibres perpendicular to the load case have resulted in a tighter standard deviation under compression and lower stiffness and strength; this may be used in the future to produce materials with engineered failure modes. In contrast, the addition of reed fibres coaxial to the load case have been shown to increase both stiffness and strength in a more balanced manner than densification. The addition of these hollow fibres contributed to a slight decrease in density, thus resulting in a more efficient performance tuning. Substrate supplementation has also been shown to increase performances without influencing density. Graphical and numerical analysis of MBC performances reports may be developed in the future so as to analyse further the role of cultivation factors upon composite behaviour.

While MBC are still in early development and necessitate further studies to become competitive to other interior or structural applications, the idea of structuring the composites may scale to architectural applications with the translation of composition strategies to using kagome. In this spirit, the structural analysis that we reported in this study probed the design space by looking at the structural capacity of kagome gridshells when applied a wet mycelium composite dead load. The use of members with higher bending stiffness, such as carbon fibre, results in a stable structure under load. Because this material is particularly environmentally detrimental, we developed a hybridisation strategy in the weave, by using carbon fibre members along the principal lines of stress and bamboo members in the remaining of it. The simulation of this case resulted in minimal displacements, it shall be prototyped in the future and other load cases will be simulated (such as snow load, punctual load, wind load).

Finally, we report on the thermal characterisation of hemp shive based and birch shaving based MBC. In the 40 and 80 mm thickness groups, the hemp substrate was approximately 11 % less conductive than birch. The birch substrate material density was XX % lower than that of the hemp material. Similarly to the difference observed between extruded polystyrene (XPS, $\lambda \in [0.029 - 0.032]$ W/mK at 25 °C for a 50 mm thickness) and expanded polystyrene (EPS, $\lambda \in [0.030 - 0.040]$ W/mK at 25 °C for a 75 mm thickness), the increase in material density pairs in our case with a densification of the mycelium present between substrate particles (we can notice the difference in packing density in Fig. 49), leading to an increase in hyphal density and a higher number of phase-change. Molecular excitation is then diffused more efficiently in the hemp substrate, resulting in a lower material conductivity. Future studies may investigate the effect of substrate chemical profile to mycelial expression, and for a normalised density, report on its effect upon conductivity. To the best of our knowledge, we report for the first time on the effect of composing buildups of MBC insulation. Although the unitary series supports the idea that mycelium can be instrumentalised for its ability to build highly redundant hyphal networks contributing to oppose energy transmission, such buildups have not been investigated yet. They may be explored in the future with the addition of low thermal conductivity materials such as ceramics.

Across this report we identified and have attempted to make the principles driving MBC design and engineering comprehensible. But a number of other wide ranging consequences of this prac-

tice regard architecture, of which sustainability. Because mycelium-based composites have a higher vernacular potential [99], they can be made of recycled substrate, or materials available locally and seasonally. MBC benefit also from a unique aesthetic as a result of the phylogenetic expression of fungi through craft and substrate design [101] (Fig. 55). We hope that the on-going uncovering of the principles for predicting MBC performance contributes to an engineering practice that may satisfy industrial needs (which actualises in patents [24]), but also fosters a rich, open, and creative citizen craft. It is not everyday that a novel, fully sustainable, and affordable material is discovered.



Figure 55: For a given fungal species, a range of more-than-visual aesthetics can be crafted.

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APPENDICES